MIL-C-60215(MU)
5 March 1965
SUPERSEDING
FA(FC)-PD-504A
1 May 1963

MILITARY SPECIFICATION

COMPUTER LOGIC UNIT: 8623272

1. SCOPE

1.1 This specification covers one type of Computer Logic Unit, 8623272 commonly called FLAT.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of
MIL-R-60176	Rectifier, Transistor Assembly
MIL-T-60177	Transistor Assembly
MIL-P-60178	Power Supply Subassembly
MIL-C-60180	Crosspoint
MIL-C-60184	Computer Set, Gun Direction: 10525738
MIL-G-60186	Generator Pulse
MIL-N-60187	Network E
MIL-D-60188	DU-DL Driver
MIL-N-60189	Network D
MIL-N-60195	Neon Driver
MIL-L-60196	Logic/Flip-Flop
MIL-R-60214	Readout Driver
MIL-V-60216	Voltage Regulator Subassembly
MIL-V-60217	Voltage Regulator Subassembly
MIL-V-60218	Voltage Regulator Subassembly
MIL-V-60219	Voltage Regulator Subassembly

FSC 1220

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F8623272	Computer Logic Unit: 8623272
F8623274	Control Box, Computer Logic Unit Test Set C-4020/GSM-70
B8623275	Tape, Test Set Computer
A10531830	Reproducer, Signal Data AN/GSQ-64

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The unit shall be manufactured in accordance with Drawing F8623272 and drawings pertaining thereto.

3.2 General specifications.- The unit shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Wiring
b. Threaded fastners
c. Marking
d. Finishes

3.3 Assemblies.- The assemblies listed below shall have been inspected and accepted in accordance with their respective drawings and specifications prior to being assembled in the FALT:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

(a)	MIL-R-60176	Rectifier, Transistor Assembly
(b)	MIL-T-60177	Transistor Assembly
(c)	MIL-P-60178	Power Supply Subassembly
(d)	MIL-C-60180	Crosspoint
(e)	MIL-G-60186	Generator, Pulse
(f)	MIL-N-60187	Network E
(g)	MIL-D-60188	DU-DL Driver
(h)	MIL-N-60189	Network D
(i)	MIL-N-60195	Neon Driver
(j)	MIL-L-60196	Logic/Flip-Flop
(k)	MIL-R-60214	Readout Driver
(l)	MIL-V-60216	Voltage Regulator Subassembly
(m)	MIL-V-60217	Voltage Regulator Subassembly
(n)	MIL-V-60218	Voltage Regulator Subassembly
(o)	MIL-V-60219	Voltage Regulator Subassembly

3.4 Environmental service conditions.

3.4.1 Shock and vibration.- The FALT, in its transit case, shall be capable of withstanding a circular motion of 1 ± 0.1 inch diameter at a frequency of 261 ± 2 cycles per minute for a total of 3 hours. The horizontal motion shall be constrained to a minimum of 3.25 inches total excursion. The unit shall not be damaged or adversely affected by this shock and vibration and subsequently shall meet all the requirements of this specification.

3.4.2 Sand and dust.- The set shall meet the requirements of 3.5 at an ambient temperature of +75 ± 15°F while subjected for 6 hours to an atmosphere of sand and dust. The density of the sand and dust shall be 0.1 to 0.5 grams per cubic foot of air with a velocity of 200 ± 100 feet per minute. The relative humidity (RH) shall not exceed 30%. The computer shall also meet the requirements of 3.5 after exposure for 6 hours to the same sand and dust conditions at +160°F.

3.4.3 Temperatures.

3.4.3.1 Operating.- The unit shall meet the requirements of 3.5.24 while exposed to temperatures of -40°F and +135°F.

3.4.3.2 Storage.- The unit shall meet the requirements of 3.5 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.4.4 Humidity.

3.4.4.1 Operating.- The unit shall meet the requirements of 3.5.24 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.4.4.2 Non-operating.- The unit shall meet the requirements of 3.5 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.4.5 Fungus resistance.- The unit shall show no evidence of damage and shall meet the requirements of 3.5, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

Source: https://assist.dla.mil -- Downloaded: 20260410-26T04:20Z
Check the source to verify that this is the current version before use.

3.4.6 Elevation.

3.4.6.1 Operating.- The unit shall be operated and meet the requirements of 3.5.24 for a period of 8 ± 1 hours while exposed to conditions encountered at 10,000 feet.

3.4.6.2 Non-operating.- The unit shall meet the requirements of 3.5 after exposure, for 8 ± 1 hours, to conditions encountered at 50,000 feet.

3.4.7 Salt laden air.- The unit shall not be damaged or its performance impaired from exposure to salt laden air as encountered in coastal areas during operation or exposure.

3.4.8 Immersion.- The unit shall be capable of continuous performance after being immersed with all covers in place and pressure relief valve closed in 3 feet head of water for 2 hours. The water temperature shall be 70°F.

3.4.9 Rain and wind (equipment non-operating).- The unit shall be capable of continuous performance after exposure to simulated rainfall of 2 inches per hour accompanied by winds at a rate of 20 mph for 4 hours. The water temperature shall be 60°F.

3.5 Performance.

3.5.1 Power supplies.

3.5.1.1 Output voltage.- When energized from either a 100 ± 4.5 volts rms or 140 ± 4.5 volts rms (line to neutral), 400 ± 20 cycles, 3 phase, 4 wire AC source, the power supplies shall meet the requirements of Table I under the min-max load conditions of Table II.

3.5.1.2 Kick-out protection circuits.- When energized with 120 ± 4.5 volts rms (line to neutral) and loaded with the max-load and kick-out loads of Table II, each of the +6, -6, -18, and ± 150 volt regulated power supplies individually shall trip to a deenergized condition; deenergizing all of the regulated supplies a 0 ± 1 volt output condition. The trip memory voltage (TKO) shall be .5 ± .5 volts for the supply which has been overloaded and +6 ± 1 volts for all other protected circuits.

3.5.1.3 Marginal test voltages.- When the power supplies are energized with 120 ± 4.5 volt rms (line to neutral), the output voltages shall be as indicated in Table III for each position of the marginal test switch.

3.5.1.4 Phase sequence.- The blowers in FALT shall provide their maximum air flow (correct directional rotation) across the main chassis when either an A-B-C or B-A-C phase sequence is used on the main power source.

3.5.2 Clock pulse.- The clock pulse shall meet the requirements of Figure I when observed at the OC and 1C pins on the pulse generator board of FALT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

3.5.3 Automatic test circuits.

(a) Set, reset, and test.- With a voltage of -18 ± 1.8V on the -18 VFT input and -6 ± .6 volts on the selection lines (listed in Table IV and V) of each flip-flop board, a clock pulse as shown in Figure 2 when fed into the CTI input of the clock board with a static clock configuration (ie: FD1 at Ground, FD7 at -6 volts and FD6 at -18 volts) shall be capable of setting each flip-flop to either of its two states depending on whether FD4 or FD5 on the clock board is at -6 ± .6 volts (the other being +3 ± .3 volts). Each flip-flop through crosspoint circuits on the flip-flop board shall be tested for both set and reset conditions.

(b) Action signal drivers and primary gate.- The outputs of primary AND gates and drivers specified in Table VI which are connected to an external connector shall display requirements of Table VI.

3.5.4 Voltage monitor lines.

3.5.4.1 Internal voltages.- The internal voltages, as per Table VII, shall read in the green portion of the voltage monitor scale. The voltages, as indicated on the voltage selector switch, engraved in yellow may read in the yellow portion of the monitor scale.

3.5.4.2 External voltages.

(a) The voltage monitor circuit shall read within the green portion of the scale when the listed nominal voltages are impressed on the pins of Table VII.

(b) The voltage monitor shall not read in the red portion of the scale when either the max. or min. listed voltages are impressed on the pins of Table VII.

(c) The voltage monitor shall read in the green portion of the scale when +6 ± 1 volt is impressed on each of the lines listed in Table VIII. The voltage monitor shall read in the red portion of the scale when 0.5V ± .5V is impressed on each line in Table VIII.

3.5.5 Crosspoint input leads.- The crosspoint input leads from external jacks J1, J2, and J3 (Pins E, F, q, r, s, t, u, v, w, x, y, z, AA, BB, CC, DD, EE, FF, GG, HH, JJ, KK, LL, MM, NN, PP) shall be properly connected, with no shorts between leads, to their crosspoint inputs.

3.5.6 Self test switches and test normal light.- The test normal light shall be energized (on) or deenergized (off) as shown in Table IX. The SFC, Test Mode, TAP, TFO, TFI, switches shall be in the positions shown in Table IX.

3.5.7 Pulse counter.- The pulse counter shall be capable of performing with the limits specified for the characteristics in 4.9.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

3.5.8 Set commands.- The set command ISA (one-set all), OSF (zero-set the following flip-flop), OSA (zero-set all), ISF (one-set the following flip-flop), and BSC (both sides clocked) shall set and reset the exciter flip-flops E1 through E8 in accordance with the procedure of 4.9.10.

3.5.9 Input register.- The input register shall be capable of performing within the limits specified for the characteristics in 4.9.11.

3.5.10 Marker register.- The marker register shall function as represented by the marker selector display within the limits specified for characteristics in 4.9.12.

3.5.11 Index counter.- The index counter shall function as represented by the index selector display within the limits specified for characteristics in 4.9.13.

3.5.12 Halt and delete commands.- The halt and delete commands shall function within the limits specified for characteristics in 4.9.14.

3.5.13 Mode control.- The mode control shall function within the limits specified for characteristics in 4.9.15.

3.5.14 Manual halt.- The manual halt mode shall function within the limits specified for characteristics in 4.9.16.

3.5.15 Program skip.- The program skip mode shall function within the limits specified for characteristics in 4.9.17.

3.5.16 Jump command.- The jump command mode shall function within the limits specified for characteristics in 4.9.18.

3.5.17 Pre-set stops.

(a) Pre-set stop at normal (command halt).- With the preset stop in the normal position, the preset-stop mode shall function in accordance with the procedure of Section 4.9.19.

(b) Pre-set stop at clock (halt at logic clock test).- With the pre-set stop switch in the clock position, the preset stop mode shall function in accordance with the procedure of Section 4.9.20.

(c) Pre-set stop at index (halt at selected index).- With the pre-set stop switch in the index position, the preset stop mode shall function in accordance with the procedure of Section 4.9.21.

3.5.18 Command entry and 0 flip-flop command loading.- Manual command entry operation shall function in accordance with the procedure of Section 4.9.22 and 4.9.23.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

3.5.19 Threshold amplifiers.

(a) The threshold amplifiers on boards 308, 309, and 310 shall be in the "on" condition as indicated by lamps F1, A_p, F_o respectively being lighted with an input voltage of -4.08 ± 02 volts and shall be in the "off" condition as indicated by the specified lamp being unlighted with an input voltages of -3.92 ± .02 volts.

(b) The threshold amplifiers on board 325 shall be in the "on" condition as indicated by a -6.8 ± .5 voltage at the output with an input of -1.83 ± .02 volts and shall be in the false conditions as indicated between zero and minus 0.25 volts at the output for an input of -1.67 ± .02 volts. An appropriate auxiliary indicator lamp may be used to show the on-off conditions at the output.

3.5.20 Error detector.- The error detector shall function within the limits specified for characteristics in 4.9.24.

3.5.21 Matrix driver output test.

(a) The DU-DL outputs at pins J4-A, B, C, D, E, F, G, H, J, L, K, M, N, P, R, S, T, U, V, W, X, Y, Z shall, when properly addressed by internal logic, be at a voltage between -5.0 volts and -7 volts and when not addressed shall be at a voltage of -.15 ± .15 volts. "When all DU-DL drivers are addressed true, the output voltage shall be between -5 volts and -7.8 volts."

(b) The DUR-DLR outputs at pins J4-a, b, c, d, e, f, g, h, shall when properly addressed by internal logic be at a voltage of between -5.0 volts and -7 volts and when not addressed shall be at a voltage of +2.85 ± .15 volts.

3.5.22 FIT and FOT lines.- The FIT and FOT lines shall function within the limits specified for characteristics in 4.9.26.

3.5.23 -18 set Voltage Command.

(a) The "-18VFT OFF" command shall reset flip-flop E18 and a voltage of 0 ± .1 volt shall appear on J4-PP.

(b) The "-18 VFT ON" command shall set flip-flop E18 and a voltage between -17.46 and -18.54 shall appear on J4-PP.

3.5.24 FALT self test.- The FALT shall function correctly when the self test tape B8623275 is used in all positions of the marginal test switch.

3.6 FALT Run-in.- The FALT shall be subjected to a 6½ hour run-in time (excluding down-time) subsequent to the successful completion of the performance requirements of 3.5 of this specification. The run-in time shall include the vibration of 3.6.1 and the temperature operation of 3.6.2. During the operational part of run-in the FALT self-test tape shall be run continuously.

3.6.1 Vibration.- The FALT shall not be damaged after being subjected to a circular motion of 1 ± 0.1 inch diameter at a frequency of 271 ± 2 cycles per minute, with horizontal motion constrained to a minimum of 3.25 inches. The FALT shall be vibrated for 15 minutes in its normal operating position, and then 15 minutes up-side down.

3.6.2 Temperature operation.- The FALT shall be operated successfully for 2 hours at +125°F, for 2 hours at -22°F, and for at least 2 hours at nominal room temperature.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables A, B, and C and the test methods and procedures in 4.9. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table A shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

TABLE A
(100 percent inspection)

	Characteristic	Requirement	Test Procedure
101.	Power supplies	3.5.1	4.9.2
102.	Clock pulse	3.5.2	4.9.3
103.	Automatic test circuits	3.5.3	4.9.5
104.	Voltage monitor lines	3.5.4	4.9.6
105.	Crosspoint input leads	3.5.5	4.9.7
106.	Self test switches and test normal light	3.5.6	4.9.8
107.	Pulse counter	3.5.7	4.9.9
108.	Set commands	3.5.8	4.9.10
109.	Input register	3.5.9	4.9.11
110.	Marker register	3.5.10	4.9.12
111.	Index counter	3.5.11	4.9.13
112.	Halt and delete commands	3.5.12	4.9.14
113.	Mode control	3.5.13	4.9.15
114.	Manual halt	3.5.14	4.9.16
115.	Program skip	3.5.15	4.9.17
116.	Jump command	3.5.16	4.9.18
117.	Pre-set stops	3.5.17	4.9.20, 4.9.21
118.	Command entry and 0 flip-flop command loading	3.5.18	4.9.22 4.9.23
119.	Threshold amplifiers	3.5.19	4.9.4
120.	Error detector	3.5.20	4.9.24
121.	Matrix driver output test	3.5.21	4.9.25
122.	FIT and FOT lines	3.5.22	4.9.26
123.	-18 set voltage command	3.5.23	4.9.27
124.	FALT self test	3.5.24	4.9.28
125.	FALT run-in	3.6	4.10

4.3 Control tests.

4.3.1 General.- One unit shall be selected at random as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table B.

TABLE B

	Control Test	Requirement	Test Procedures
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.3.2 Environmental.- Three units shall be selected at random as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table A and shall then meet the requirements and tests in Table C.

TABLE C

	Control Test	Requirement	Test Procedures
303.	Shock and vibration	3.4.1	4.5
304.	Sand and dust	3.4.2	4.6
305.	Operating temperature	3.4.3.1	4.7.1
306.	Storage temperature	3.4.3.2	4.7.2
307.	Humidity - operating	3.4.4.1	4.8.1
308.	Humidity - Non-operating	3.4.4.2	4.8.2
309.	Fungus	3.4.5	4.8.6
310.	Elevation - operating	3.4.6.1	4.7.3
311.	Elevation - Non-operating	3.4.6.2	4.7.4
312.	Salt laden air	3.4.7	4.8.3
313.	Immersion	3.4.8	4.8.4
314.	Rain and Wind - Non-operating	3.4.9	4.8.5

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.9.

Item Number	Description
1. Volt/Ohmmeter	A D.C. range at least to 200 volts at no less than 10,000 ohms/volt. A.C. voltage at least to 300 volts (rms) at no less than 5000 ohms/volt. Capable of measuring 0 to 100 megohms. Voltage accuracies of at least ±3% of full scale (A.C. voltage of 60 cps). Ohmmeter accuracy of at least ±3% of D.C. scale with full battery.
2. Resistance Box; 6 Decade	1,000,000 ohm total in 1 ohm steps.
3. Voltmeter, D.C. Digital	A range of at least .001 to 999 volts D.C., or three significant figures. Accuracy of at least ± 0.2% of reading ± 1 count. Input impedance of at least 10 megohms on all ranges.
4. Photo-Reader	Capable of accepting and reading the five channel FALT tapes at a rate of 500 to 700 words/second. The output voltage shall be -12 ± .5V for a true signal, and ground for a false signal. Pulse duration shall be between 0.45 and 0.25 milliseconds. It shall produce a strobe pulse of between 0.85 and 0.98 millisecond duration, with +6 ± .5 volts for a true signal and -12 ± .5V for a false signal. It shall be able to transmit a "system ready" (TR) voltage of -12 ± .5 volts.
5. Control Box, Computer Logic Unit, Test Set C-4020 ( ) GSM-70	
6. Oscilloscope	Dual channel, rise time of scope plus measuring probe 50 milli-microseconds or less, sweep rate of 50 milli-microseconds per centimeter or greater, input resistance no less than 8 megohms, input capacitance less than 25 pico-farads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.5 Shock and vibration.- The FALT in its transit case shall be placed on a test table, but not fastened to it. Wooden fences shall be placed around the case and shall constrain its horizontal motion to a maximum of 3.25 inches total excursion. The equipment shall be vibrated at 271 ± 2 cycles per minute for a total of 3 hours. During each 1/2 hour period the transit case, containing the FALT, shall rest on a different face. The motion of all points on the table shall be circular with a 1.0 ± 0.1 inch diameter. During the test the FALT shall not be operated. It shall be subsequently inspected for physical damage, and shall be subjected to the tests defined in 4.9 of this specification.

4.6 Sand and dust.- This test shall be conducted in accordance with the sand and dust test defined in MIL-F-13926. The test shall not be repeated at 0°F, and temperature changes shall not occur at a rate exceeding 2°F per minute. During the operating portion of the test, and subsequent to the non-operating tests, the FALT air filters shall be replaced as required to obtain proper operation. Subsequent to the tests the unit shall be inspected for physical damage, and shall be subjected to the tests defined in 4.9 of this specification.

4.7 Temperature and elevation tests.

4.7.1 Operating temperatures.- This test shall be conducted in accordance with Procedure I, Conditions C and E of the temperature tests defined in MIL-F-13926 with the following exceptions and additions:

(a) The temperature shall never be changed at a rate exceeding 2°F per minute.

(b) After temperature stabilization occurs at +125°F, the unit shall remain non-operating at that temperature for 64 ± 4 hours, and shall then be operated for 8 ± 1 hours. Subsequent to operation, the unit shall be non-operating while the temperature is returned to room ambient.

(c) After temperature stabilization occurs at -40°F, the unit shall be energized and, after the "Power Ready" lamp comes on, shall be operated for 8 ± 1 hours. The unit shall be non-operating while the temperature is returned to room ambient.

(d) Throughout the operating portions of this test the unit shall be subjected to the tests defined in 4.9.28 of this specification, and shall meet the requirements specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.7.2 Storage temperatures.- This test shall be conducted in accordance with Procedure II, Conditions A and H of MIL-F-13926. With the following exceptions and additions:

(a) The temperature shall never be changed at a rate exceeding 2°F per minute.

(b) The FALT covers shall be properly in place throughout the test, and shall be removed as required for performance of subsequent operating tests.

(c) After temperature stabilization at -80°F occurs, the temperature shall be maintained constant for a period of 24 ± 2 hours. The temperature shall then be returned to room ambient and temperature stabilization shall occur before FALT operation is initiated.

(d) After temperature stabilization at +160°F occurs, the temperature shall be maintained constant for 4 hours. The temperature shall then be returned to room ambient and temperature stabilization shall occur before FALT operation is initiated.

(e) Subsequent to the extreme temperature tests, the unit shall be inspected for physical damage, and shall be subjected to the tests defined in 4.9 of this specification.

4.7.3 Elevation - Non-operating.- The unit with covers in place shall be placed within a test chamber and the absolute internal pressure of the chamber reduced at 3.44 inches of mercury (corresponding to an altitude of 50,000 feet above sea level) and an ambient temperature of -65°F. The rate of change of pressure shall not exceed 2.44 inches of mercury per minute, and the rate of change of temperature 2°F per minute. The pressure and temperature shall be maintained constant for 8 ± 1 hours and shall be returned to room nominal conditions at the same rates. Subsequent to exposure, the unit shall be inspected for physical damage, and shall be subjected to the tests defined in 4.9 of this specification.

4.7.4 Elevation - operating.- The FALT shall be placed within a test chamber and the absolute internal pressure of the chamber reduced to 20.58 inches of mercury (corresponding to an altitude of 10,000 feet). The rate of change of pressure shall not exceed 2.44 inches of mercury per minute. The temperature shall be maintained at 75 ± 15°F. The unit shall be operated for 8 ± 1 hours in accordance with 4.9.28 of this specification. The return to room ambient pressure shall be at a maximum of 2.44 inches of mercury per minute.

4.8 Moisture resistance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.8.1 Humidity - Non-operating.- This test shall be conducted in accordance with the humidity test defined in Procedure I of MIL-F-13926, except that the 4-1/2 hour period shall be reduced to 2 hours, and the 6 hour period reduced to 2 hours and the test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and subjected to the test defined in 4.8.2 of this specification and shall meet the specified requirements.

4.8.2 Humidity - operating.- This test shall be conducted in accordance with the humidity test defined in Procedure III of MIL-F-13926, except that the upper temperature shall be modified to 125°F, the humidity shall be maintained at 92 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202B shall be excluded). Throughout the test the unit shall be operated in accordance with 4.9.28 of this specification, and shall meet the specified requirements.

4.8.3 Salt laden air.- The unit shall be subjected to salt laden air in accordance with the following procedure:

(a) The FALT covers shall be in place during non-operating periods, and shall be removed only as required during operating tests.

(b) The unit shall be operated for 8 hours and stored for 16 hours for 4 cycles (32 hours of operation).

(c) Throughout the test the unit shall be exposed to an atmosphere equivalent to that which would prevail at an ocean shore within 100 yards of the high tide mark. Direct salt spray or salt fog need not be present.

(d) The operating temperatures shall be within the range of 0 to +100°F, and the non-operating temperatures within the range of -10 to +110°F.

(e) The unit shall be subsequently inspected for physical damage, and shall be operated for 4 hours at room ambient conditions. Operation during and subsequent to the test shall be in accordance with 4.9.28 of this specification.

4.8.4 Immersion.- FALT, with all covers in place, and pressure relief valve closed, shall be immersed under a 3 feet ± 2 inch head of water for a period of 2 ± 1/10 hours. The covers shall then be removed and FALT shall be operated in accordance with 4.9.28 of this specification. The water temperature shall be 70 ± 10°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.8.5 Rain - Non-operating rain.- FALT, with all covers in place, shall be exposed to simulated rainfall at the rate of 2 ± 1/2 inches per hour, accompanied by a wind at a rate of 20 ± 2 miles per hour for a period of 4 ± 1/2 hours. The wind shall be directed at the front surface of the front cover of FALT, and be inclined 45 ± 20 degrees from the horizontal. FALT shall be mounted on the field table in its normal position. After exposure to this environment, covers shall be removed and FALT shall be subjected to the test defined in 4.9.28 of this specification. The water temperature shall be 60 ± 25°F.

4.8.6 Fungus.- This test shall be conducted in accordance with the fungus test defined in MIL-F-13926.

4.9 Electrical performance.

4.9.1 The tests of 4.9.2 that require removal of circuit boards need only be performed once for a single machine. Any test which calls for a repeat of 4.9 can start at 4.9.2(a).

4.9.2 Power supplies.- For all power supply tests the FALT switches shall be set as per Table X, unless otherwise specified. The FALT power supplies shall be tested with all printed circuit subassemblies disconnected. Connect phases A, B, and C of the 3 phase 120 ± 4.5 volt rms, line to neutral supply to J11-A, B, and C respectively. Connect neutral to J11-D. With the main power circuit breakers off, measure the voltage from CBOA-2, CBOB-2 and CBOC-2 to neutral. The voltage shall be 0 volts.

With the circuit breaker switches on, apply +150 ± 5 volts D.C. to relay contacts K1-4, and ground at K1-11. Press the power on button. Measure the line to neutral voltage at CBOA-2, CBOB-2 and CBOC-2. It shall be 120 ± 4.5 volt rms.

Using Item I (4.4.1.1), measure the A.C. voltages between phases AB, BC, and CA of the TR-1, 2, and 3 secondaries. The voltages and measurements shall be as indicated in Table XXII. Connect printed circuit subassemblies Regulator 1, Regulator 2, Regulator 3, Regulator 4, Network E, Diode Rectifier, Transistor and Capacitor boards into the system. Tests, operation, and measurements shall be in accordance with the following procedures:

(a) Connect the resistors of column C, Table II to the indicated supplies at the terminal block in the FALT main frame. With A.C. line adjusted to 100 ± 4.5 volts rms, perform the measurements of Table I. Repeat the measurements with the line adjusted to 140 ± 4.5 volt rms.

(b) Replace the resistors at the terminal block with the resistors of column B, Table II. Repeat the measurements of Procedure (a) above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

(c) With the line adjusted to 120 ± 4.5 volt rms, connect a resistor from column D, Table II, across the supply and in parallel with the resistor of column B. Observe that all regulated supplies have de-energized, and that the output voltages are 0 ± 1 volt except the +150R output voltage shall be less than +40 volts when the maximum load is applied (with the supply in KO state) and less than +12 volts with the maximum load and kickout load in parallel. Re-apply FALT power and sequentially repeat the procedure using each resistor of column D across the indicated power supply. As each resistor is connected, the associated TKO voltage as read on the voltage monitor circuit shall read low and in the "red" portion of the voltmeter. All other TKO voltages shall read center scale in the "green" portion of the voltmeter. For this measurement, the unit selector switch shall be in the FALT position and the positive or negative voltage selectors shall be used to select the particular TKO.

TABLE I

A SUPPLY	B D.C. VOLTAGE TO GROUND	C MAX (MV) RIPPLE*	D MEASURED FROM	D MEASURED TO(GND)
+6R	+6 ± 0.12	50	318-T06	320-T05
+3R	+3 ± 0.12	50	320-T14	320-T06
-3R	-3 ± 0.12	50	318-T01	320-T06
-10R	-10 ± 0.2	50	316-T09	320-T05
+150R	+150 ± 2.32	500	218-T02	216-T11
+15R	+15 ± 0.45	50	318-T09	314-T06
-1.5R	-1.5 ± 0.045	20	314-T03	318-T19
+1.25R	+1.25 ± 0.038	100	320-T12	318-T21
-18R	-18 ± 0.54	500	320-T02	318-T21
+35R	+35 ± 0.7	50	314-T09	316-T06
-6R	-6 ± 0.18	300	318-T24	318-T19

* Ripple is defined as the peak-to-peak value of any alternating component of the output voltage. The ripple voltage measurement shall exclude noise of less than 10 microseconds duration. The noise level measurement of less than 10 microseconds duration shall be no greater than ± 0.75 volts off of the D.C. level.

(d) Set the "Marginal Test" switch to each position listed in Table III and measure the output voltages. Readings shall be as indicated in Table III. The confidence switch shall be in the off position for each of these readings and shall remain off for all subsequent tests unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

(e) Phase sequence test.- With power on, observe that the blowers are provided an air flow into the FALT main chassis across the boards. Reverse the phase sequence of the 208V, 400 cycle, 3 phase power input by reversing two of the phase leads. Ovserve that the blower rotates in the same direction as before. Depress and hold the pushbutton switch located on the FALT main frame (S 50) until one of the squirrel cage blowers stops rotation. Release the switch and the fan shall start again.

TABLE II

A	B	C	D
SUPPLY	MAX LOAD OHMS ± 5% TO GROUND	MIN LOAD OHMS ± 5% TO GROUND	KICKOUT LOAD OHMS ± 5% TO GROUND
+6R	20, 5 watt	60, 2 watt	2, 25 watt
+3R	3.1, 5 watt	10.3, 2 watt	
-3R	9.23, 5 watt	None	
-10R	15.4, 25 watt	54, 5 watt	
+150R	1185, 5 watt	5200, 2 watt	330, 100 watt
+15R	107, 25 watt	277, 25 watt	
-1.5R	1.25, 12 watt	6.8, 1 watt	
+1.25R	1.21, 5 watt	3, 5 watt	
-18R	4.7, 150 watt	12.7, 150 watt	1.8, 150 watt
+35R	305, 50 watt	None	
-6R	1.61, 50 watt	7.52, 50 watt	0.5, 100 watt

TABLE III

NOMIN SUPPLY VOLT	MARGINAL SWITCH POSITION						MEASURED	
	OFF	#1	#2	#3	#4	#5	FROM	TO(GND)
-18	-18	-17.1	-18.9	-17.1	-17.1	-18.9	320-T02	318-T21
-10	-10	-9.5	-10.5	-9.5	-9.5	-10.5	316-T09	320-T05
-6	-6	-5.7	-6.3	-6.3	-5.7	-6.3	318-T24	318-T19
+1.25	+1.25	+1.19	+1.31	+1.31	+1.19	+1.19	320-T12	318-T21
+6	+6	+6.3	+5.7	+5.7	+5.7	+5.7	318-T06	320-T05
+15	+15	+15.75	+14.25	+15.75	+14.25	+15.75	318-T09	314-T06
+35	+35	+33.25	+36.75	+33.25	+36.75	+33.25	314-T09	316-T06

NOTE: All D.C. Voltages are ± 2%
A.C. Source = 120 ± 4.5 Volt RMS, 400 ± 20 CPS

4.9.3 Clock pulse.- All printed wiring assemblies shall be connected for the following tests. With the "Test Mode" switch in normal, observe the clock pulse at J209 A5 and J209 B-13 using Item 9 (oscilloscope). Ground is at J209 A-1. Refer to Figure 1. The signal shall have the following characteristics:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the current version before use.

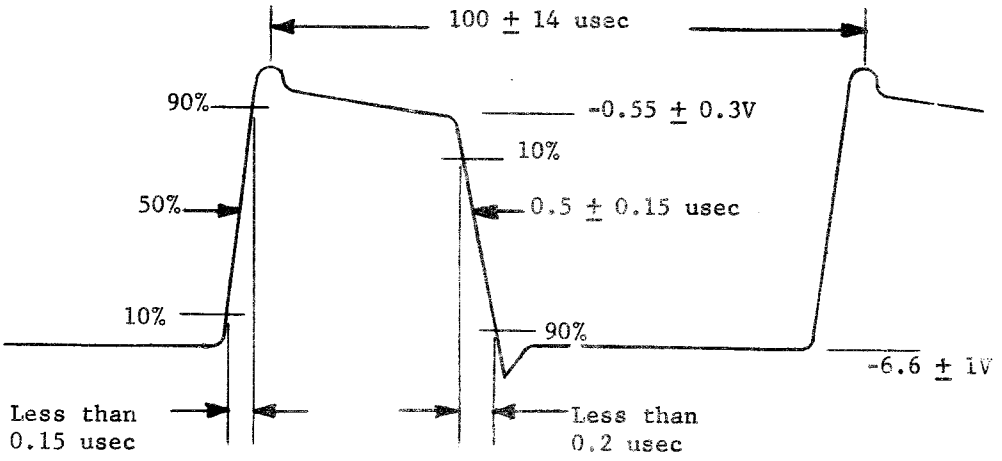


FIGURE 1

(a) The rise time and fall time of the clock pulse, when measured at each pin specified, shall be not greater than 0.15 microseconds and 0.2 microseconds, respectively, at the 10% and 90% points.

(b) Pulse width at the 50% level, when measured at the point specified, shall be between 0.35 and 0.65 microseconds.

4.9.4 Threshold adjustments.

Step 1 Open the FALT to provide accessibility to the threshold adjustment potentiometers on boards 308, 309, 310, and 325.

Step 2 Turn FALT power on. Place the TFI switch in the "ON" position. The F1 lamp shall light.

Step 3 Connect Item 2, Resistance Box, and Item 3, Digital Voltmeter from the TFI jack on the front panel to ground.

Vary the setting of the resistance box, to obtain each of the following conditions.

(1) The F1 lamp out for a voltage reading on the digital voltmeter which lies between 3.90 and 3.94, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

(2) The F1 lamp lighted for a voltage reading which lies between 4.06 and 4.10.

Step 4 If it is not possible to attain the conditions of step 3, adjust the resistance box until the digital voltmeter indicates a voltage of -4.0 ± .02 volts. Adjust potentiometer R30 on the network "D" board in J308 until the F1 lamp dims or flickers. Repeat step 3.

Step 5 Connect Item 1 (VOM) to pin J325-49A. An auxiliary indicator and driver may be used in place of the VOM.

Step 6 Vary the setting of the resistance box to obtain each of the following conditions.

(1) The voltage on the VOM reading -.2 ± .2 volts (or the auxiliary indicator being unlighted) for a voltage on the digital voltmeter between 1.65 and 1.69 volts.

(2) The voltage on the VOM reading -6.8 ± .5 (or the auxiliary indicator being lighted) for a voltage on the digital voltmeter between 1.81 and 1.85 volts.

Step 7 If it is not possible to attain the conditions of step 6, adjust the resistance box until the digital voltmeter reads -1.75 ± .02 volts. Adjust potentiometer R6 on the neon driver board in J325 until the VOM flickers (or the auxiliary indicator dims or flickers) between -6.8 ± .5 and -.2 ± .2. Repeat step 6.

Step 8 Place the TFI switch off and the TFO switch on. Connect the resistance box, and digital voltmeter from the TFO jack on the front panel to ground and adjust the resistance box to obtain the on-off conditions of step 3 for the FO lamp.

Step 9 If it is not possible to attain the conditions of step 8, adjust the resistance box until the digital voltmeter reads -4.0 ± .02 volts. Adjust potentiometer R30 on the network "D" board in J310 until the FO lamp dims or flickers. Repeat step 8.

Step 10 Connect the VOM (or the input of the auxiliary driver) to pin J325-29A. Repeat step 6.

Step 11 If it is not possible to attain these conditions, repeat step 7 but make the adjustment on R4. Repeat step 10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 12 Place the TFO switch "off" and the TAP switch "on". Connect the resistance box and digital voltmeter from the TAP jack on the front panel to ground, and adjust the resistance box, to obtain the on-off conditions of step 3 for the AP lamp.

Step 13 If it is not possible to attain the conditions of step 12, adjust the resistance box until the digital voltmeter reads -4.0 ± .02 volts. Adjust potentiometer R30 on network "D" board in J309 until the AP lamp dims or flickers. Repeat step 12.

Step 14 Connect the VOM (or the input of the auxiliary driver) to pin J325-64A. Repeat step 6.

Step 15 If it is not possible to attain these conditions, repeat step 7 but make the adjustment on R8. Repeat step 14.

4.9.5 Automatic test circuits.

(a) A voltage of not greater than -16.2 and not less than -19.8 volts shall be applied to pin PP of J10 (-18V FT). Connect -6 ± 0.12 volts to pin J10-G and -18 ± 0.36 volt to pin J10-X. Ground pins J10-L and J10-M with the power supplies.

(b) Zero-set all (OSA).- Prior to making a one-set test on a given flip-flop, the following initialization procedure shall be performed to insure all flip-flops are in zero state:

(1) Apply an input level of between -6.6 and -5.4 volts to each of the selection lines listed in Table IV.

(2) Apply an input level of between 2.7 and 3.3 volts to the one inhibit F (FD4) line of pin J10-J.

(3) Apply an input level of between -6.6 and -5.4 volts thru a 1K ohm resistor to the zero inhibit F (FD5) line on pin J10-K.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

TABLE IV

A	B
Selection Line	J10 Pin #
DUZ1	A
DUY1	B
DUZ2	C
DUY2	D
DUZ3	E
DUY3	F
DL1Z	N
DL2Z	P
DL3Z	R
DL4Z	S
DL5Z	T
DL6Z	U
DL7Z	V
DL8Z	W

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

TABLE V

A	B	C	A	B	C
Flip-Flop	Selection Lines		Flip-Flop	Selection Lines	
E18	DUZ3	DL8Z	R63	DUY2	DL7Z
R61	DUZ3	DL7Z	R11	DUZ2	DL7Z
R62	DUY3	DL5Z	R12	DUY2	DL8Z
M1	DUY1	DL2Z	R13	DUZ2	DL8Z
M2	DUZ1	DL2Z	R14	DUY3	DL1Z
M3	DUY1	DL4Z	R21	DUZ3	DL1Z
M4	DUZ1	DL4Z	R22	DUY3	DL2Z
O1	DUY3	DL6Z	R23	DUZ3	DL2Z
O2	DUY1	DL5Z	R24	DUY3	DL3Z
O3	DUZ1	DL5Z	R31	DUZ3	DL3Z
P1	DUY1	DL6Z	R32	DUY3	DL4Z
P2	DUZ1	DL6Z	R33	DUZ3	DL4Z
P3	DUY1	DL7Z	R34	DUY3	DL5Z
P4	DUZ1	DL7Z	R41	DUZ3	DL5Z
R1	DUY2	DL3Z	R42	DUY1	DL8Z
R2	DUZ2	DL3Z	R43	DUZ1	DL8Z
R3	DUY2	DL4Z	R44	DUY2	DL1Z
R4	DUZ2	DL4Z	R51	DUZ2	DL1Z
R6	DUY2	DL5Z	R52	DUY2	DL2Z
R7	DUZ2	DL5Z	R53	DUZ2	DL2Z
R8	DUY2	DL6Z	R54	DUY3	DL7Z
R9	DUZ2	DL6Z	V	DUZ3	DL6Z

NOTE: Above tables used for tests in 4.9.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

TABLE VI

A	B	C	D	E
Term	Initial-ization	Set-up	Flip-Flops	Pin
/ATR/	OSA (Zero-set all	1SF (One-set the following flip-flops	M4, M2	J10-H
"	1SA (One-set all	-	-	"
RLQ	OSA	1SF	R4	J10-Y
"	1SA	OSF (Zero set the following flip-flops	R3	"
RLR	OSA	1SF	R4, R3	J10-Z
"	1SA	- -	-	"
RLS	OSA	1SF	R3	J10-a
"	1SA	OSF	R4	"
/RLT/	OSA	-	-	J10-b
"	1SA	OSF	R4, R3	"
/RLU/	OSA	1SF	R2	J10-c
"	1SA	OSF	R1	"
/RLV/	OSA	1SF	R2, R1	J10-d
"	1SA	-	-	"
RLW	OSA	1SF	R1	J10-e
"	1SA	OSF	R2	"
RLX	OSA	-	-	J10-f
"	1SA	OSF	R2, R1	"
RUQ	OSA	1SF	R9	J10-g
"	1SA	OSF	R8	"
RUR	OSA	1SF	R9, R8	J10-h
"	1SA	-	-	"
RUS	OSA	1SF	R8	J10-k
"	1SA	OSF	R9	"
RUT	OSA	-	-	J10-m
"	1SA	OSF	R9, R8	"
RUUO	OSA	1SF	R7	J10-n
"	1SA	OSF	R6	"
/RLA/	OSA	1SF	R4, R2	J10-p
"	1SA	OSF	R3, R1	"
RUVO	OSA	1SF	R7, R6	J10-w
"	1SA	-	-	"
/RLC/	OSA	1SF	R4, R1	J10-x
"	1SA	OSF	R3, R2	"
RUWO	OSA	1SF	R6	J10-y
"	1SA	OSF	R7	"
RLB	OSA	1SF	R4, R2, R1	J10-DD
"	1SA	OSF	R3	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

TABLE VI (Cont'd)

A	B	C	D	E
Term	Initial-ization	Set-up	Flip-Flops	Pin
RUXO	OSA	-	-	J10-EE
"	1SA	OSF	R7, R6	"
RLE	OSA	1SF	R4, R3, R2	J10-FF
"	1SA	OSF	R1	"
/CTO/	OSA	1SF	P2, P3, P4, M2, M1, 01	J10-GG
"	1SA	OSF	M3, 03, 02, P1	"
/CTO/	OSA	1SF	P2, P3, P4, M2, M1, 03, 01	"
"	1SA	OSF	P1, M3	"
1HO	OSA	1SF	P4, P2, M1, 01	J10-HH
"	1SA	OSF	P1, M3, 02, 03	"
1H1	OSA	1SF	P4, P2, M1, 01, 03	J10-JJ
"	1SA	OSF	P1, M3, 02	"
/OSR2/	OSA	1SF	P4, E18, P3, P2, M1, M2, 01, 03	J10-KK
"	1SA	OSF	P1, M3, 02	"
/SR2/	OSA	1SF	P4, P3, P2, M1, M2, 01, E18	J10-LL
"	1SA	OSF	P1, M3, 03, 02	"
/HTR/	OSA	-	-	J10-NN
"	1SA	OSF	M2, M1	"

NOTE: The above table used for tests in 4.9.5.

(4) Apply an input pulse equivalent to that shown in Figure 2 to the Clock Trigger, F(CTO), line on Pin J10-MM.

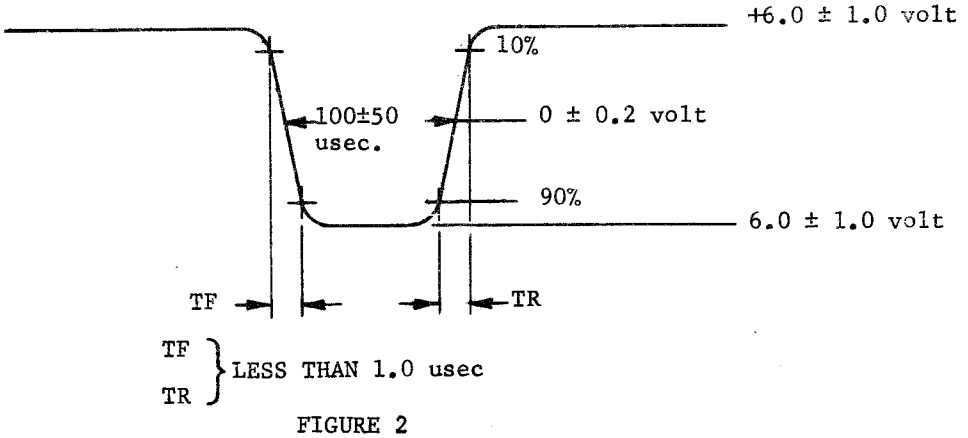


FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.

(c) One-set all.- Prior to making a zero-set test on a given flip-flop, the following initialization procedure shall be performed to insure all flip-flops are in the one state:

(1) Apply an input level of between minus 6.6 and -5.4 volts to each of the selection lines listed in Table IV.

(2) Apply an input level of between 2.7 and 3.3 volts to the Zero Inhibit, F(FD5), line on Pin J10-K.

(3) Apply an input level of between -6.6 and -5.4 volts thru a 1K OHM resistor to the one inhibit, F(FD4), line on Pin J10-J.

(4) Apply an input pulse equivalent to that shown in Figure 2 to the Clock Trigger, F(CTO) line on Pin J10-MM.

(d) One-set tests.- Each of the flip-flops shall be tested for one-set operation by energizing the appropriate internal automatic test diodes. The zero-set all initialization procedure specified in paragraph above shall be performed prior to any individual one-set test. The following test procedure shall be performed for each of the flip-flops listed in Table V.

(1) Apply an input level between minus 6.6 and minus 5.4 volts to the two selection lines associated with the flip-flop under test as listed in Table V.

(2) Apply a level of between minus 0.6 and +0.6 volts to each of the selection lines not associated with the flip-flop under test.

(3) Perform the operations specified in paragraphs c(2), (3), and (4) above.

(4) Output measured on the one sense ($^{FF}1T$) line on Pin J10-i shall be between -6.5 and -4.5V.

(5) Output measured on the zero sense (^{F}FOT) line at Pin J10-j shall be between -1 and +3.0 volts.

(e) Zero-set tests.- Each of the flip-flops shall be checked for zero-set operation by energizing appropriate internal automatic test diodes. The one-set all initialization procedure specified in paragraph c above shall be performed prior to any individual zero-set test. The following test procedure shall be performed for each of the flip-flops listed in Table V.

(1) Apply an input level of between -6.6 and -5.4 volts to the two selection lines associated with the flip-flop under test, as listed in Table V.

(2) Apply a level of between -0.6 and +0.6 to each of the selection lines not associated with the flip-flop under test.

(3) Perform the operation specified in paragraphs b(2), (3), and (4) above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

(4) Output measured on the zero sense (FFOT) line at Pin J10-j shall be between -6.5 and -4.5 volts.

(5) Output measured on the one sense (FFIT) line at Pin J10-i shall be between -1.0 and +3.0 volts.

(f) Logic output.- The automatic test output lines listed in Table VI are energized by combinations of the outputs of computer flip-flops. Each of the outputs shall be tested by the following test procedures:

(1) Perform the initialization procedure of paragraphs 4.9.5(b) and 4.9.5(c) specified in Column B of Table VI for the output line under test.

(2) For those cases where set-up is specified in Columns C and D, Table VI, the output voltage measured on the line under test shall be between -1 and +0.5 volts except the output voltage for the following terms: The output voltage for 1HO and 1HI shall be -6.2 ± 0.2 volts, for /CTO/ and /HTR/ shall be + 6 ± 1 volts, and for /OSR2/ and /1SR2/ shall be + 2.84 ± .26 volts. For those cases where no set-up is required, perform the measurements specified in paragraph 4.9.5(f)(4).

(3) Using the procedures of paragraphs d(1) through (4) above, and e(1) through (4) above, set-up flip-flops associated with the output line under test as specified in Columns C and D of Table VI.

(4) Output voltage measured on the line under test shall be between -6.5 and -4.5 volts except the output voltage for the following terms: The output voltage for IHO and 1HI shall be 2.84 ± 0.26 volts, and for /CTO/, /OSR2/, /1SR2/, /ATR/, and /HTR/ shall be -6 ± 1 volts.

4.9.6 Voltage monitor lines.

4.9.6.1 External voltages.

(a) Voltage monitor lines shall be tested by impressing the nominal voltage on each of the pins listed in the Table VII. The unit selector switch shall be placed in external and the positive voltage selector and negative voltage selector, respectively, positioned to the voltage to be read. The voltage monitor meter shall read within the green portion of the scale for each voltage.

(b) Voltage monitor lines shall be tested by impressing both the maximum and minimum voltages listed on each of the pins of Table VII. The voltage monitor meter shall not read in the red portion of the scale for each voltage selected.

(c) The TKO voltage monitor lines listed in Table VIII with their respective pins shall be tested by impressing first + 6 ± .5 VDC on each. The positive or negative voltage selector then shall be positioned to the particular KO voltage line being measured. The voltage monitor shall read in the green portion of the meter. A voltage of .5V ± .5V then shall be impressed on each KO voltage line. The voltage monitor shall then read low in the red portion of the meter scale.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.6.2 Internal voltages.- Turn power on and place the voltage monitor switch in FALT position. Set the appropriate voltage selector switch to each of the voltages of Table VII. As a voltage is selected, the voltage monitor meter shall read in the green portion of the scale. Those voltages on the selector switch which are engraved in yellow may read in the yellow portion of the scale.

TABLE VII

Nominal Voltage	Pin	Maximum Voltage	Minimum Voltage
-6	J3 - B	- 6.30	- 5.70
-18	" - M	-18.9	-17.1
+35	" - P	+36.8	+33.2
+6	" - b	+ 6.6	+ 5.3
+20	" - c	+21.0	+19.0
-27	" - d	-28.4	-25.6
-32	" - e	-33.6	-30.4
-10	" - f	-10.5	- 9.50
-12	" - g	-12.6	-11.4
-3	" - h	-3.30	- 2.70
-1.5	" - i	- 1.65	- 1.35
+1.25	" - j	+ 1.38	+ 1.12
+3	" - k	+ 3.30	+ 2.70
+6 (z)	" - m	+ 6.30	+ 5.70
+15	" - n	+15.8	+14.2
+150	" - p	+158	+142

NOTE: All voltages specified shall have a ±1% tolerance when applied as per 4.9.6.

TABLE VIII

TKO VOLTAGE	PIN
+35KO	J3-U
+15KO	J3-V
+ 6KO	J3-W
- 6KO	J3-X
-10KO	J3-Y
-18KO	J3-Z
+150KO	J3-a

NOTE: The above tables are used for tests in 4.9.6.

4.9.7 Crosspoint input leads test for shorts.

(a) Remove boards 224, 225, 226, 227, 308, 309, 310, 210.

(b) Test all pins of J2 and all pins of J1 except A, B, C, S, and g. Test the leads of J3 (listed below) for correct wiring to their terminal points on J224, J225, J226, J227, J308, J309, J310, and J210. There shall be no shorts

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

between any of these leads when individually tested from any one lead to all others:

PINS	PINS	PINS
E	y	MM
F	z	NN
q	AA	PP
r	BB	
s	CC	
t	DD	
u	EE	
v	FF	
w	GG	
x	HH	
	JJ	
	KK	
	LL	

(c) Close and secure the FALT.

4.9.8 Self-test switches and "test normal" light.

Refer to Table IX. Place switches in Columns 1, 2, 3, 4, and 5, in their listed positions. "Test normal" lamp located on front panel will be effected as indicated in Column 6.

TABLE IX

1	2	3	4	5	6
SFC	Test Mode	TAP	TFO	TFI	Test Normal
Normal	Normal	Off	Off	Off	On
Test	Normal	Off	Off	Off	Off
Normal	Static Clock	Off	Off	Off	Off
Normal	Normal	On	Off	Off	Off
Normal	Normal	Off	On	Off	Off
Normal	Normal	Off	Off	On	Off
Normal	Normal	Off	Off	Off	On

4.9.8.1 Logic test set-up.

Step 1 Refer to Table X. Place the switches listed in Column 1 in the positions listed in Column 2.

TABLE X

1	2
Switches	Positions
Operate	MAN
Program Skip "A"	OFF
Program Skip "B"	OFF

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

TABLE X (Cont'd)

1	2
Switches	Positions
Program Skip "C"	OFF
Delay Start	OFF
Preset Stop	NORM
Homing	OFF
Index Selectors	000
Marker Selectors	001
Test Mode	STATIC CLOCK
SFC	NOR
Program	MAN ENTRY
TFI	OFF
TFO	OFF
TAP	OFF
Homing Select	Marker/Index
WR SW ENABLE	OFF
Confidence Test	OFF
Marginal Test	OFF
Neg. Voltage Selector	OFF
Pos. Voltage Selector	OFF
Unit Selector	OFF

Step 2 Connect Manual Control Box (Item 5) to J12 of FALT. Place switch #5 of manual control box in down (OFF) position.

Step 3 Using the volt/ohm meter (Item 1), measure the dc voltage at the 1C and OC test point with respect to the "GRD" test point. The voltage measured shall lie between the limits of -7.2 and -5.8 volts.

Step 4 Connect the oscilloscope (Item 6) between the 1C test jack on the front panel, and ground. Operate the clock key on the manual entry box once, and as the key is depressed observe a single pulse on the scope consistent with Figure 1 except that the pulse width shall be 1.6 ± .5 microseconds. Repeat the test at the OC test jack.

4.9.9 Pulse counter test.

Step 1 (a) Push and hold the power "ON" button.
*(b) Operate the "Clock" key three times.

*NOTE: The phrase (Operate the "Clock" Key) shall be interpreted to mean depress and then release the "Clock" key located on the Manual Control Box.

(c) Release the "Power On" button.
(d) The "PH" and "Marker/Index" lamps shall light.
(e) The following lamps shall be unlighted:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Operate
Program Skip "A"
Program Skip "B"
Program Skip "C"
Homing "HS"
Delay Start "DS"
Error
Test Normal
Board Readout
All lights on manual entry box except R5.

Step 2 Repeatedly operate the "Clock" key until lamps P_1, P_2, P_3, and P_4, are out.

Step 3
(a) Place and hold the "Hand Sprocket" switch on the Manual Control Box in the "OPERATE" position.
(b) Operate the "Clock" key twice.
(c) The P_1 lamp shall be lighted.
(d) The P_2, P_3, and P_4 lamps shall not be lighted.
(e) Release the "Hand Sprocket" switch.
(f) Operate the "Clock" key. Lamp P4 shall light.
(g) Operate the "Clock" key. Lamp P1 shall go out.
(h) Operate the "Clock" key. Lamp P4 shall go out; lamp P3 shall light.
(i) Operate the "Clock" key. Lamp P3 shall go out.

Step 4
(a) Lamps R1 through R9 on the Manual Control Box shall not be lighted.
(b) Lamps P1 through P4 on the FALT front panel shall not be lighted.
(c) Place switch #5 on the Manual Control Box in the up position. Manual Control Lamp "5" shall light.
(d) Place and hold the "Hand Sprocket" switch in "OPERATE" position.
(e) Operate the "Clock" key twice; the P1 and P2 lamps shall light in that sequence.
(f) Release the "Hand Sprocket" switch. Leave "Manual Control Switch "5" locked in the up position.

Step 5
(a) Operate the "Clock" key. Lamp P4 shall light.
(b) Operate the "Clock" key. Lamp P1 shall go out; lamp P3 shall light.
(c) Operate the "Clock" key. Lamp P3 shall go out.
(d) Operate the "Clock" key. Lamp P4 shall go out.
(e) Operate the "Clock" key. Lamps P1, P2, and P3 shall be lighted. Lamp P4 shall be unlighted.
(f) Operate the "Clock" key. Lamps P1, P2, P3 and P4 shall not change state.
(g) The PH lamp on the FALT front panel shall be lighted. This indicates that the M flip-flops are all zero-set.

Step 6
(a) Depress and hold the "Power On" button. Operate the "Clock" key. Lamp P1 shall go out. Release the "Power On" button.
(b) Operate the "Clock" key. Lamp P2 shall go out.
(c) Depress and hold the "Power On" button. Operate the "Clock" key. Lamp P4 shall light. Release "Power On" button.
(d) Operate the "Clock" key. Lamp P1 and P2 shall light. Lamps P3 and P4 shall remain lighted.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.

(e) Operate the "Clock" key. Lamp P2 shall go out.
(f) Operate the "Clock" key. Lamps P1 and P3 shall go out.
(g) Operate the "Clock" key twice. All lamps P1, P2, P3, and P4 shall go out. This shows that the P flip-flops are zero-set.
(h) Repeat Steps 4.d. through 5.g.
(i) Place and hold the "Hand Sprocket" switch in the clear position.
(j) Operate the "Clock" key.
(k) Release the "Hand Sprocket" switch.
(l) Operate the "Clock" key twice. Lamps P1, P2, P3 and P4 shall be out.
(m) Repeat Steps 4.d. through 5.g.
(n) Place and hold the "Operate" switch in the "RESET" position.
(o) Operate the "Clock" key.
(p) Release the "Operate" switch.
(q) Operate the "Clock" key twice. Lamps P1, P2, P3 and P4 shall be out.

Step 7
(a) Depress and hold "Manual Control" switch "1". Operate the "Clock" key. Release "Manual Control" switch "1". "Manual Control" lamp "1" shall light.
(b) Depress and hold "Manual Control switch "2". Operate the "Clock" key. Release "Manual Control" switch "2". "Manual Control" lamp "2" shall light.
(c) Depress and hold "Manual Control" switch "3". Operate the "Clock" key. Release "Manual Control" switch "3". "Manual Control" lamp "3" shall light.
(d) Depress and hold "Manual Control" switch "4". Operate the "Clock" key. Release "Manual Control" switch "4". "Manual Control" lamp "4" shall light.

Step 8
(a) Place and hold the "Operate" switch on the FALT front panel in the "RESET" position.
(b) Operate the "Clock" key once. Release the "Operate" switch to the "MANUAL" position.
(c) Lamp P3 shall light. Operate the "Clock" key once.
(d) "Manual Control" lamps "1", "2", "3", and "4" shall go out.
(e) Lamp P3 shall go out.

Step 9
(a) Depress and hold switch "1" on Manual Control Box.
(b) Operate the "Clock" key.
(c) Release "Manual Control" switch "1".
(d) Repeat Steps a., b., and c. using "Manual Control" switch "2".
(e) Repeat Steps a., b., and c. using "Manual Control" switch "3".
(f) Repeat Steps a., b., and c. using "Manual Control" switch "4".
(g) Place and hold the "Hand Sprocket" switch in "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch.
(j) The following lamps shall light.

Manual Control "1"	Manual Control "4"
Manual Control "2"	Manual Control "5"
Manual Control "3"	P1
	P2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

(k) Operate the "Clock" key. Lamps P1, P2, and P4 shall light.

Step 10
(a) Operate the "Clock" key. Lamps P2, P3, and P4 shall light.
(b) Operate the "Clock" key. Lamps P2 and P4 shall light.
(c) Operate the "Clock" key. Lamp P2 shall light.
(d) Operate the "Clock" key. Lamps P1, P2, and P3 shall light
(e) Operate the "Clock" key. Lamps P2 and P3 shall light.
(f) Operate the "Clock" key. Lamps P3 shall light.
(g) Operate the "Clock" key. The following lamps shall not be lighted: P1, P2, P3, P4 - Manual Control #'s 1, 2, 3, and 4.

Step 11
(a) Depress and hold "Manual Control" switch "1".
(b) Operate the "Clock" key.
(c) Release "Manual Control" switch "1".
(d) Repeat Steps a., b., and c. using "Manual Control" switch "2".
(e) Repeat Steps a., b., and c. using "Manual Control" switch "4".
(f) Repeat Steps a., b., and c. using "Manual Control" switch "6". The Command Halt lamp shall light.
(g) The "Preset Stop" switch shall be in the "NORMAL" position.
(h) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(i) Operate the "Clock" key twice.
(j) Release the "Hand Sprocket" switch.
(k) Operate the "Clock" key.
(l) The following lamps shall be lighted: P1, P2, P4, R1, R2, R4, R and R6.
(m) The R3 lamp shall not be lighted.

Step 12
(a) Operate the "Clock" key.
(b) Lamps P1 and P4 shall go out.
(c) Lamp P3 shall light. The PH lamp shall light.
(d) Operate the "Clock" key.
(e) Lamps P_1 and P_3 shall be lighted.
(f) Lamps P_2 and P_4 shall be out. The PH lamps shall remain lighted.
(g) Operate the "Clock" key.
(h) The following lamps shall remain lighted: R_1, R_2, R_4, and PH.
(i) The following lamps shall not be lighted: P_1, P_2, P_3, and P_4.

Step 13
(a) Place and hold the "Hand Sprocket" switch in the "CLEAR" position until Step 13h.
(b) Operate the "Clock" key.
(c) The following lamps shall not be lighted: P_1, P_2, and P_4.
(d) The following lamps shall be lighted: P_3, R_1, R_2, and R_4.
(e) Operate the "Clock" key.
(f) The following lamps shall not be lighted: P_1, P_2, P_3, R_1, R_2, R_4 and CH.
(g) The P4 lamp shall be lighted.
(h) Release the "Hand Sprocket" switch.
(i) Operate the "Clock" key twice.
(j) The following lamps shall not be lighted: P1, P2, P3, P4, R1, R2, R4 and CH.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 14
(a) Depress and hold the "Power On" button and operate the "Clock" key until the "Marker Selectors" display is blank. Release the "Power On" button.
(b) Place and hold the "Operate" switch in the "Reset" position and operate the "Clock" key twice.
(c) Release the "Operate" switch and operate the "Clock" key until lamps P1, P2, P3, and P4 are not lighted.
(d) Turn "Manual Control: switch "5" down; it will lock. Lamps 5 shall go out.
(e) Depress and hold "Manual Control" switch "1".
(f) Operate the "Clock" key. Release switch "1".
(g) Place and hold the "Hand Sprocket" switch in "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch.
(j) Operate the "Clock" key.
(k) The following lamps shall light:

P1
P4
"Manual Control "1"
OPERATE

Step 15
(a) Operate the "Clock" key.
(b) Lamp P1 shall go out.
(c) Operate the "Clock" key.
(d) Lamp P4 shall go out. "Manual Control" lamp "6" shall light. Lamp P3 shall light.
(e) Operate the "Clock" key.
(f) Lamp P3 shall go out. "Manual Control" lamp "1" shall go out.

Step 16
(a) Depress and hold "Manual Control" switch "2".
(b) Operate the "Clock" key. Release "Manual Control" switch "2".
(c) Depress and hold the "Hand Sprocket" switch in "OPERATE" position.
(d) Operate the "Clock" key twice.
(e) Release the "Hand Sprocket" switch.
(f) Operate the "Clock" key.
(g) The following lamps shall be lighted:

P1
P4
Manual Control "2"
Manual Control "6"

(h) Operate the "Clock" key.
(i) Lamp P3 shall light. Lamp P1 shall go out. Lamp P4 shall remain lighted.
(j) Operate the "Clock" key.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

(k) The following lamps shall be lighted:

P1	P3
P2	P4
Manual Control "2"	
Manual Control "6"	

(l) The "Marker Selector" display shall read ##1.*

*NOTE: The number sign (#) shall be used in marker and index display readings to indicate that a nixie tube displays no number whatsoever.

Step 17
(a) Operate the "Clock" key four times.
(b) "Manual Control" lamp "7" shall light. The "Homing" lamp shall light.
(c) The following lamps shall go out:

P1	Manual Control "2"
P2	Manual Control "3"
P3	Manual Control "4"
P4	Manual Control "5"
Manual Control "1"	Manual Control "6"

Step 18
(a) Place and hold "Hand Sprocket" switch in clear position.
(b) Operate "Clock" key four times.
(c) Release "Hand Sprocket" switch.
(d) "Manual Control" lamp "7" shall go out.
(e) Depress and hold "Manual Control" switch "3".
(f) Operate the "Clock" key.
(g) Release "Manual Control" switch "3".
(h) Repeat Steps e., f., g., using "Manual Control" switch "4".
(i) "Manual Control" lamps "3" and "4" shall be lighted.
(j) Place and hold the "Hand Sprocket" switch in the "Operate" position.
(k) Operate the "Clock" key.
(l) Release the "Hand Sprocket" switch.
(m) Operate the "Clock" key four times.
(n) "Manual Control" lamps "3" and "4" shall go out and "Manual Control" lights "8" and "9" shall be lighted.
(o) Repeat Steps 18j through 18m.
(p) "Manual Control" lamps "8" and "9" shall go out.
(q) Push the operate switch to reset and operate the clock key until the operation light comes on. Release the operate switch.

4.9.10 Commands set-reset tests.- ALL VOLTAGES MEASURED UNDER THIS TEST SHALL BE REFERRED TO POWER GROUND.

NOTE: Hereinafter "Manual Control" switches "1", "2", "3", "4", "5", "6", "7", "8" and "9" shall be referred to as switches "1", "2", "3", "4", "5", "6", "7", "8" and "9".

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 1 Zeroset all (OSA) to establish initial conditions.

(a) Set 5 in the up position.
(b) Depress and hold switch "9".
(c) Operate the "Clock" key. Lamp "9" shall light.
(d) Release switch "9".
(e) Lamps "5" and "9" shall be lighted.
(f) Place and hold "Hand Sprocket" switch in "Operate" position.
(g) Operate "Clock" key twice.
(h) Release "Hand Sprocket" switch.
(i) Operate "Clock" key eight times.
(j) The following lamps shall not be lighted:

E1 through E8 and P1 through P4.

(k) Switch "5" shall remain in "up" position. Lamp "5" shall remain lighted.

Step 2 One-set all (ISA) Command Test.

(a) Depress and hold switch "9".
(b) Operate the "Clock" key. Lamp "9" shall light.
(c) Release switch "9".
(d) Repeat Steps 2.a through 2.c for switch "1".
(e) Lamps "1", "5", and "9" shall be lighted.
(f) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(g) Operate the "Clock" key twice.
(h) Release the "Hand Sprocket" switch.
(i) Lamps P1 and P2 shall be lighted. Lamps P3 and P4 shall not be lighted.
(j) The following voltages shall be measured.*

J10-GG	+5	to	+7
"-JJ	-5	to	-7
"-HH	-5	to	-7
"-LL	+2.58	to	+3.1
"-KK	+2.58	to	+3.1

(k) Refer to Table XI for the remaining procedure for Step 2.

Step 3 Zero-set flip-flop (OSF) E1 test.

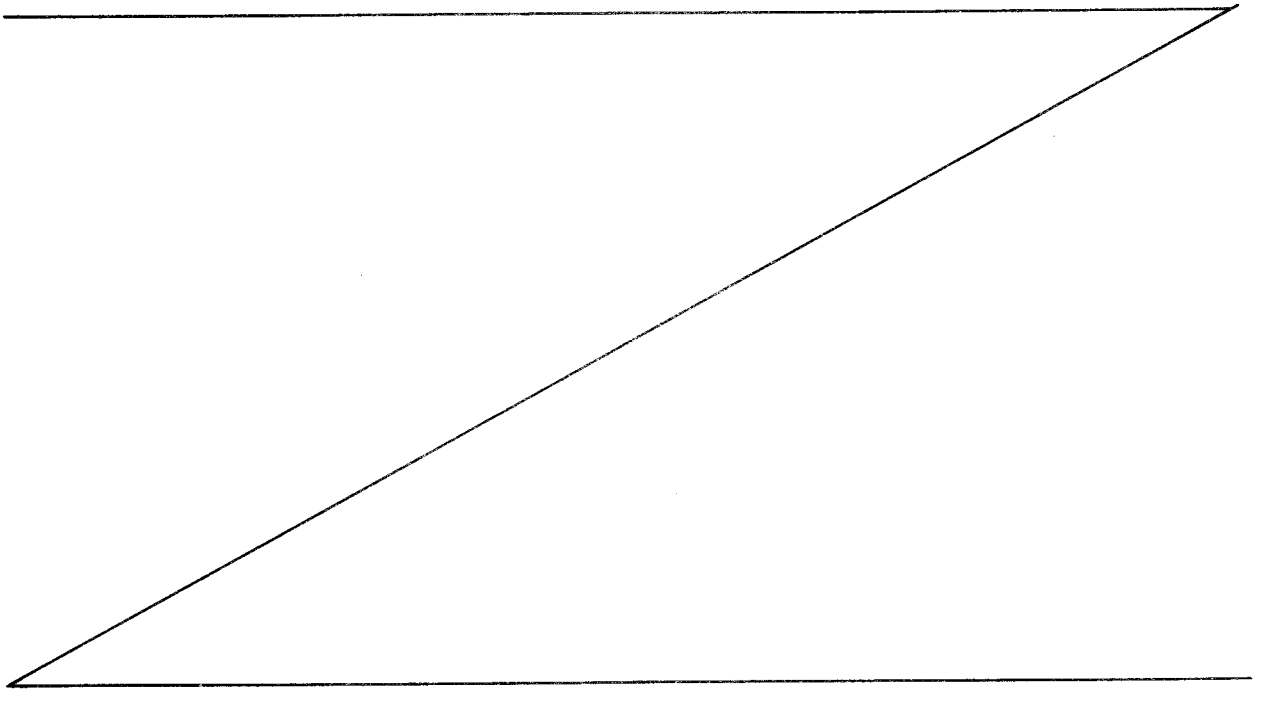
(a) Depress and hold switch "9".
(b) Operate the "Clock" key. Lamp 9 shall light.
(c) Release switch "9".
(d) Depress and hold switch "6".
(e) Operate the clock key. Lamp 6 shall light.
(f) Release switch 6.
(g) Depress and hold switch 4.
(h) Operate the "Clock" key. Lamp 4 shall light.
(i) Release switch 4.
(j) Depress and hold switch "2".
(k) Operate the "Clock" key. Lamp 2 shall light.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 3 Cont'd

(l) Release switch "2".
(m) Depress and hold switch "1".
(n) Operate the "Clock" key. Lamp 1 shall light.
(o) Release switch "1".
(p) Hold the hand sprocket switch in the operate position.
(q) Operate the "Clock" key twice.
(r) Release the "Hand Sprocket" switch.
(s) Lamps E1 through E8 shall remain lighted. Lamps P1 and P2 shall be lighted.
(t) Repeat the voltage measurements of step 2j.
(u) Refer to Table XII for the remaining procedure for Step 3.

*NOTE: The following procedure may be followed in lieu of the voltage measurements described in this section. Connect indicator lights between each of pins J10-GG, JJ, HH, LL, KK and ground. The lights shall work at the voltage specified in this section, turning on or off with a change in state of the circuit being monitored. Make an initial measurement of the prescribed voltages of 2j at the voltage extremes encountered in circuit operation. After correct light operation is assured, the indicator lights may be used to denote the state of the driver rather than a voltage measurement.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.10 Step 2 (cont'd)

TABLE XI

	Operate the "Clock" key	Lamp Conditions		Using the Item I read the following:									
		Lighted	Unlighted	CTO Voltage on Pin J10-GG		1H1 Voltage on Pin J10-JJ		1HO Voltage on Pin J10-HH		ISR Voltage on Pin J10-LL		OSR Voltage on Pin J10-KK	
				Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
l.	once	P1, P4, P2 OP, ISF	P3	+5.58	+6.1	-7	-5	-7	-5	+2.58	+3.1	+2.58	+3.1
m.	once	P2, P3, P4, OP, ISF	E1 to E8 P1	-7	-5	-7	-5	+2.58	+3.1	-7	-5	+2.58	+3.1
n.	once	E1, E2, E3, E4, E5, E6, E7, E8, P2, P4, ISF	P1, P3 OP	+5.58	+6.1	-7	-5	+2.58	+3.1	+2.58	+3.1	+2.58	+3.1
o.	once	E1,E2,E3, E4,E5,E6, E7,E8,P2, ISF	P1, P3, P4, OP										
p.	once	E1,E2,E3, E4,E5,E6, E7,E8,P1, P2,P3,OP, ISF	P4										
q.	once	E1,E2,E3, E4,E5,E6, E7,E8,P2, P3,OP,OSF	P1, P4										
r.	twice	E1,E2,E3, E4,E5,E6, E7,E8,OP	P1, P2, P3, P4, PH										

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.10 Step 3 (cont'd)

TABLE XII

	Operate the "Clock" key	Lamp Conditions		Using the Item I read the following:									
		Lighted	Unlighted	CTO Voltage on Pin J10-GG		IHI Voltage on Pin J10-JJ		IHO Voltage on Pin J10-HH		ISR Voltage On Pin J10-LL		OSR Voltage On Pin J10-KK	
				Min.	Max.	Min.	Max	Min	Max	Min	Max	Min	Max
k.	once	E8 E1,E2,E3, E4,E5,E6, E7,P1,P2, P4,OSF OP	P3	+5.58	+6.1	-7	-5	-7	-5	+2.58	+3.1	+2.58	+3.1
l.	once	E1,E2,E3, E4,E5,E6, E7,P2,P3, P4, OSF OP E8	P1	-7	-5	+2.58	+3.1	-7	-5	+2.58	+3.1	-7	-5
m.	once	E2,E3,E4 E5,E6,E7, P2, P4, OSF E8	P1, P3, E1 OP	+5.58	+6.1	+2.58	-3.1	-7	-5	+2.58	+3.1	+2.58	+3.1
n.	once	E2,E3,E4, E5,E6,E7, P2, OSF E8	E1, P1, P3, P4, OP										
o.	once	E2,E3,E4, E5,E6,E7, P1,P2,P3, OP,OSF E8	P4, E1										
p.	once	E2,E3,E4, E5,E6,E7, P2,P3,OP, OSF E8	P1, P4, E1										
q.	twice	E2,E3,E4, E5,E6,E7, OP, OSF E8	P1, P2, P3, P4, E1										

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.10 Step 4 (Cont'd)

TABLE VIII

Operate the "Clock" key	Lamp Conditions		Using the Item I read the following:									
	Lighted	Unlighted	CTO Voltage on Pin J10-GG		IHI Voltage on Pin J10-JJ		IHO Voltage on Pin J10-HH		ISR Voltage on Pin J10-LL		OSR Voltage on Pin J10-KK	
			Min.	Max	Min.		Min	Max	Min	Max	Min	Max
once	P1,P2,P4, E2,E3,E4, E5,E6,E7, ISF E8 OP	P3, E1	+5.58	+6.1	-7	-5	-7	-5	+2.58	+3 1	+2.58	+3.1
once	E2,E3,E4, E5,E6,E7, P2,P3,P4, OP,ISF E8	P1, E1	-7	-5	-7	-5	+2.58	+3.1	-7	-5	+2.58	+3.1
once	E2,E3,E4, E5,E6,E7, P2,P4,E8, ISF	P1, P3, E1 OP	+5.58	+6.1	-7	-5	+2.58	+3.1	+2.58	+3.1	+2.58	+3.1
once	E2,E3,E4, E5,E6,E7, P2, E8 ISF	P1, P4, E1, P3 OP										
once	E2,E3,E4, E5,E6,E7, P1,P2,P3, OP, ISF E8	P4, E1										
three times	E2,E3,E4, E5,E6,E7, OP, ISF E8	P1, P2, P3, P4, E1										

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.10 Step 5 (Cont'd)

TABLE XIV

	Operate the "Clock" key	Lamp Conditions		Using the Item I read the following:									
		Lighted	Unlighted	CTO Voltage on Pin J10-GG		IHI Voltage on Pin J10-JJ		IHO Voltage on Pin J10-HH		ISR Voltage on Pin J10-LL		OSR Voltage on Pin J10-KK	
				Min	Max	Min	Max	Min	Max	Min.	Max	Min	Max
i.	once	E7,E2,E3, E1, P3 E4,E5,E6, E8,P1,P2, P4, OP ISF		+5.58	+6.1	-7	-5	-7	-5	+2.58	+3.1	+2.58	+3.1
j.	once	E2,E3,E4 P1 E1 E5,E6,E7, P2,P3,P4, E8, OP, ISF		-7	-5	-7	-5	+2.58	+3.1	-7	-5	+2.58	+3.1
k.	once	E1,E2,E3, P1, P3 E4,E5,E6, OP E7,P2,P4, ISF E8		+5.58	+6.1	-7	-5	+2.58	+3.1	+2.58	+3.1	+2.58	+3.1
l.	once	E2,E3,E4, P1, P3, E5,E6,E7, P4, OP E8, E1, P2, ISF											
m.	once	E1,E2,E3, P4 E4,E5,E6, E7,E8,P1, P2,P3,OP, ISF											
n.	three times	E1,E2,E3, P1, P2, E4,E5,E6, E7,E8,OP, ISF											

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 4 One-set flip-flop (ISF) tests.

(a) Depress and hold switch "9".
(b) Operate the "Clock" key. Lamp "9" shall light.
(c) Release swtich "9".
(d) Repeat Steps 4a. through 4c. for switches "2" and "1".
(e) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(f) Operate the "Clock" key twice.
(g) Release the "Hand Sprocket" switch.
(h) Lamps E2 through E8 shall remain lighted. Lamp E1 shall not light.
(i) Refer to Table XIII for the remaining procedure for Step 4.

Step 5 One-set test using the E1 flip-flop.

(a) Depress the hold switch "9".
(b) Operate the "Clock" key. Lamp "9" shall light.
(c) Release switch "9".
(d) Repeat steps 5a, b, and c for switches "6", "4", "2", and "1".
(e) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(f) Operate the "Clock" key twice.
(g) Release the "Hand Sprocket" switch.
(h) Lamps E2 through E8 shall remain lighted. Lamp E1 shall not light.
(i) Refer to Table XIV for the remaining procedure for Step 5.

Step 6 OSA Command test.

(a) Depress and hold switch "9".
(b) Operate the "Clock" key. Lamp "9" shall light.
(c) Release switch "9".
(d) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(e) Operate the "Clock" key twice.
(f) Release the "Hand Sprocket" switch.
(g) Lamps E1 through E8 shall remain lighted.
(h) Refer to Table XV for the remaining procedure for Step 6.

Step 7 (ISF), E2 test.

(a) Depress and hold switch "9".
(b) Operate the "Clock" key. Lamp "9" shall light.
(c) Release switch "9".
(d) Repeat Steps 7.a through 7.c for switches 7,4,2, and 1.
(e) Place and hold the."Hand Sprocket" switch in the "OPERATE" position.
(f) Operate the "Clock" key twice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.10 Step 6 (Cont'd)

TABLE XV

	Operate the "Clock" key	Lamp Condition		Using the Item I read the following:									
		Lighted	Unlighted	CTO Voltage on Pin J10-GG		IHI Voltage on Pin J10-JJ		IHO Voltage On Pin J10-HH		ISR Voltage on Pin J10-LL		OSR Voltage on Pin J10-KK	
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
i.	once	E1,E2,E3, E4,E5,E6, E7,E8,P1, P2,P4,OP, OSF	P3	+5.58	+6.1	-7	-5	-7	-5	+2.58	+3.1	+2.58	+3.1
j.	once	P2,P3,P4, OP, OSF, E1,E2,E3, E4,E5,E6 E7,E8	P1	-7	-5	+2.58	+3.1	-7	-5	+2.58	+3.1	-7	-5
k.	once	P2,P4,OSF	E1,E2,E3, E4,E5,E6, E7,P1,P3, OP E8	+5.58	+6.1	+2.58	+3.1	-7	-5	+2.58	+3.1	+2.58	+3.1
l.	once	P2, OSF	E1,E2,E3, E4,E5,E6, E7,P1,P3, P4, OP E8										
m.	once	P1,P2,P3 OP, OSF	E1,E2,E3, E4,E5,E6, E7,P4,E8										
n.	once	P2,P3,OP, ISF	E8,E1,E2, E3,E5,E5, E6,E7,P1, P4										
o.	twice	OP, ISF	E1,E2,E3, E4,E5,E6, E7,P1,P2, P3,P4,E8										

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 7 Cont'd.

(g) Release the "Hand Sprocket" switch.
(h) Lamps E1 through E8 shall not be lighted.
(i) Refer to Table XVI for the remaining procedure for Step 7.

Step 8 OSF Command Test.

(a) Depress and hold switch "9". Operate the "Clock" key. Release "9".
(b) Repeat 8a. for switch 2.
(c) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(d) Operate the "Clock" key twice.
(e) Release the "Hand Sprocket" switch.
(f) Lamps E1, E4, E3, E5, E6, E7, and E8 shall not be lighted. Lamp E2, P1, and P2 shall be lighted.
(g) Refer to Table XVII for the remaining procedure for Step 8.

Step 9 Zero-set test using the E2 flip-flop.

(a) Depress and hold switch "9".
(b) Operate the "Clock" key. Lamp "9" shall light.
(c) Release switch "9".
(d) Repeat Steps 9.a. through 9.c. for switch 7,4,2, and 1.
(e) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(f) Operate the "Clock" key twice.
(g) Release the "Hand Sprocket" switch.
(h) Lamp E2 shall be lighted. Lamps E1, E5, E3, E4, E6, E7, and E8, shall not be lighted.
(i) Refer to Table XVIII for the remaining procedure for Step 8.

Step 10 BSC Command test.

(a) Depress and hold switch "9".
(b) Operate the "Clock" key. Lamp 9 shall light.
(c) Release switch "9".
(d) Repeat steps 10a. through 10c. for switches "4" and "2".
(e) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(f) Operate the "Clock" key twice.
(g) Release the "Hand Sprocket" switch.
(h) Lamps E1, E2, E3, E5, E5, E6, E7, E8, P3, and P4 shall not be lighted. Lamps P1 and P2 shall be lighted.
(i) Refer to Table XIX for the remaining procedure of step 10.

4.9.11 Input register tests.

Step 1 (a) Place the "Test Mode" switch on the front of FAIT in the "NORM" Position. The Test normal lamp shall be on.
(b) Depress and release the "Power on" button.
(c) All lamps shall be out except for the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.10 Step 7 (Cont'd)

TABLE XVI

	Operate the "Clock" key	Lamp Condition		Using the Item 1 read the following							
		Lighted	Unlighted	CTO Voltage on Pin J10-GG		IHI Voltage on Pin J10-JJ		IHO Voltage on Pin J10-HH		ISR Voltage on Pin J10-LL		OSR Voltage on Pin J10-KK	
				Min.	Max.	Min	Max	Min	Max	Min	Max	Min	Max
j.	once	P1,P2,P4, OP, ISF	E1,E2,E3, E4,E5,E6, E7,P3,E8	+5.58	+6.1	-7	-5	-7	-5	+2.58	+3.1	+2.58	+3.1
k.	once	P2,P3,P4, OP, ISF	E1,E2,E3, E4,E5,E6, E7,P1,E8	-7	-5	-7	-5	+2.58	+3.1	-7	-5	+2.58	+3.1
l.	once	E2,P2,P4, ISF	E1,E3 Thru E8,P1,P3, OP	+5.58	+6.1	-7	-5	+2.58	+3.1	+2.58	+3.1	+2.58	+3.1
m.	once	E2,P2, ISF	E1,E3,E4, E5,E6,E7, P1,P3,P4, OP E8										
n.	once	E2,P1,P2, P3,OP,ISF	E1,E3,E4, E5,E6,E7, P4,E8										
o.	three	E2,OP, ISF	E1,E3,E4, E5,E6,E7, P1,P2,P3, P4,E8										

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.10 Step 8 (Cont'd)

TABLE XVII

	Operate the "Clock" key	Lamp Condition		Using the Item I read the following:									
		Lighted	Unlighted	CTO Voltage on Pin J10-GG		IHI Voltage on Pin J10-JJ		IHO Voltage on Pin J10-HH		ISR Voltage on Pin J10-LL		OSR Voltage on Pin J10-KK	
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
h.	once	E2,P1,P2, P4,OP,OSF	E1,E3,E4, E5,E6,E7, P3,E8	+5.58	+6.1	-7	-5	-7	-5	+2.58	+3.1	+2.58	+3.1
i.	once	E2,P2,P3, P4,OP,OSF	E1,E3,E4, E5,E6,E7, P1,E8	-7	-5	+2.58	+3.1	-7	-5	+2.58	+3.1	-7	-5
j.	once	E2,P2,P4, OSF	E1,E3,E4, E5,E6,E7, P1,P3,E8 OP	+5.58	+6.1	+2.58	+3.1	-7	-5	+2.58	+3.1	+2.58	+3.1
k.	once	E2,P2,OSF	E1,E3,E4, E5,E6,E7, P1,P3,P4, OP E8										
l.	once	E2,P1,P2, OP,OSF, P3	E1,E3,E4, E5,E6,E7, P4,E8										
m.	once	E2,P2,P3, OP,OSF	E1,E3,E4, E5,E6,E7, P1,P4,E8										
n.	twice	E2,OP, OSF	E1,E3,E4, E5,E6,E7, P1,P2,P3, P4,E8										

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.10 Step 9 (Cont'd)

TABLE XVIII

	Operate the "Clock" key	Lamp Condition		Using the Item I read the following									
		Lighted	Unlighted	CTO Voltage on Pin J10-GG		IHI Voltage on Pin J10-JJ		IHO Voltage on Pin J10-HH		ISR Voltage on pin J10-LL		OSR Voltage on Pin J10-KK	
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
j.	once	E2,P1,P2, P4,OP,OSF	E1,E3,E4, E5,E6,E7, P3,E8	+5.58	+6.1	-7	-5	-7	-5	+2.58	+3.1	+2.58	+3.1
k.	once	E2,P2,P3, P4,OP,CSF	E1,E3,E4, E5,E6,E7, P1,E8	-7	-5	+2.58	+3.1	-7	-5	+2.58	+3.1	-7	-5
l.	once	P2,P4, OSF	E1,E2,E3, E4,E5,E6, E7,P1,P3 E8 OP	+5.58	+6.1	+2.58	+3.1	-7	-5	+2.58	+3.1	+2.58	+3.1
m.	once	P2, OSF	E1,E2,E3, E4,E5,E6, E7,P1,P3, P4,OP E8										
n.	once	P1,P2,P3, OP,OSF	E1,E2,E3, E4,E5,E6, E7,P4,E8										
o.	three times	OP,OSF	E1,E2,E3, E4,E5,E6, E7,P1,P2, P3,P4 E8										

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.10 Step 10 (Cont'd)

TABLE XIX

	Operate the "Clock" key	Lamp Condition		Using the Item I read the following									
		Lighted	Unlighted	CTO Voltage on Pin J10-GG		IHI Voltage on Pin J10-JJ		IHO Voltage on Pin J10-HH		ISR Voltage on Pin J10-LL		OSR Voltage on Pin J10-KK	
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
j.	once	P1,P2,P4, OP,BSC	E1,E2,E3, E4,E5,E6, E7,P3,E8	+5.58	+6.1	-7	-5	-7	-5	+2.58	+3.1	+2.58	+3.1
k.	once	P2,P3,P4, OP,BSC	E1,E2,E3, E4,E5,E6, E7,P1 E8	-7	-5	-7	-5	-7	-5	+2.58	+3.1	+2.58	+3.1
L.	once	P2,P4, BSC	E1,E2,E3, E4,E5,E6, E7,P1,P3 E8 OP	+5.58	+6.1	-7	-5	-7	-5	+2.51	+3.1	+2.58	+3.1
m.	five times	OP, BSC	E1,E2,E3, E4,E5,E6, E7,P1,P2, P3,P4 E8										

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 1 (Cont'd)

(1) "Index Selectors" display shall read 000.
(2) The PH lamp shall be on.
(3) At most, one of the following "Command Indicator" lamps shall be lighted. ITF, OTF, ITA, ISF, OSF, OTA, BSC.
(4) The following lamp states are not fixed: AP, F1, FO.
(5) Test Normal.
(6) Marker/Index.

Step 2 (a) Depress switched "1" through "9" individually; the corresponding lamp for the pertinent switch shall light.

Step 3 (a) Place the "Operate" switch in the "RESET" position. Release the "Operate" switch to the "MAN" position.
(b) Lamps "1", "2", "3", "4", "6", "7", "8" and "9", shall be off. Lamp "5" shall be on.
(c) Place switch "5" down; it shall lock. Lamp "5" shall go out.
(d) Depress and release switch "1".
(e) Place the "Hand Sprocket" switch in "OPERATE" position. Release the "Hand Sprocket" switch.
(f) Lamp "6" shall be "on". Lamps "1", "2", "3", "4", "5", "7", "8", and "9" shall be out.
(g) Depress and release the "Power On" button.
(h) Lamps "1" through "9" shall be out.

Step 4 (a) Depress and release switch "2".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamp "7" shall light. Lamps "1", "2", "3", "4", "5", "6", "8", and "9" shall be out.
(d) Depress and release the "Power On" button.
(e) Lamps "1" through "9" shall be out.

Step 5 (a) Depress and release switch "3".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamp "8" shall light. Lamps "1", "2", "3", "4", "5", "6", "7", and "9" shall be out.
(d) Depress and release the "Power On" button.
(e) Lamps "1" through "9" shall be out.

Step 6 (a) Depress and release switch "4".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamp "9" shall light. Lamps "1" through "8", shall be out.
(d) Depress and release switch "8".
(e) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.12 Marker Register tests.

Step 1
(a) Depress and release switch "6".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamps "1" through "9" shall all be out.
(d) The "Marker Selector" display shall read ##1.

Step 2
(a) Depress and release switch "7".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read #12.

Step 3
(a) Depress and release switches "6" and "7".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read #123.

Step 4
(a) Depress and release switch "8".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read 234.

Step 5
(a) Depress and release switches "6" and "8".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read 345.

Step 6
(a) Depress and release switches "7" and "8".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read 456.

Step 7
(a) Depress and release switches "6", "7", and "8".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read 567.

Step 8
(a) Depress and release switch "9".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read 678.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 9 (a) Depress and release switches "6" and "9".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read 789.

Step 10 (a) Depress and release switches "7" and "9".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read 890.

Step 11 (a) Depress and release switches "6" and "9".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read 909.

Step 12 (a) Depress and release switch "9".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) Lamps "1" through "9" shall be out.
(d) The "Marker Selector" display shall read 098.
(e) Depress and release the "Power On" button. The "Marker Selector" display shall no longer be lighted.

4.9.13 Index counter tests.

Step 1 (a) Place switch "5" up; it will lock. Lamp "5" shall light.
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) The "Index Selector" display should read 001.

Step 2 (a) Place the "Hand Sprocket" switch in the "CLEAR" position, then in the "OPERATE" position. Release the "Hand Sprocket" switch.
(b) The "Index Selector" display shall read 002.
(c) Place the "Hand Sprocket" switch in the "CLEAR" position, then in the "OPERATE" position. Release the "Hand Sprocket" switch.
(d) The "Index Selector" display shall read 003.
(e) Continue repeating Step a. The "Index selector" display shall count 004, 005, etc., through 100 to 199.
(f) Place the "Hand Sprocket" switch in the "CLEAR" position, then in the "OPERATE" position. Release the "Hand Sprocket" switch.
(g) The "Index Selector" display shall read 100 again.
(h) Depress and release the "Power On" button.
(i) The "Index Selector" display shall read 000.

4.9.14 Preliminary "RUP" command tests.

Step 1 (a) Place the "Hand Sprocket" switch in the "OPERATE" position, then in the "CLEAR" position. Release the "Hand Sprocket" switch.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.

Step 1 (Cont'd)

(b) The "Index Selector" display shall read 001.
(c) Depress the release switched "1", "2", "3", "4" and "6". Switch "5" shall remain locked in the up position.
(d) Place the "Operate" switch in the "RESET" position. Release the "OPERATE" switch.
(e) Lamps "1", "2", "3", "4", "6", "7", "8", and "9" shall be off. Lamp "5" shall be lighted.
(f) The "Index Selector" display shall read 001.

Step 2 (a) Depress and release "1", "2", "3", "4" and "6".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position, then in the "CLEAR" position. Release the "Hand Sprocket" switch.
(c) The "Index Selector" display shall read 001.

Step 3 (a) Depress and release switched "1", "2", "4" and "6".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(c) The "Index Selector" display shall read 0001.
(d) The "CH" lamp shall light. The "PH" lamp shall light.
(e) Place the "Hand Sprocket" switch in the "CLEAR" position. Release the "Hand Sprocket" switch.
(f) The "CH" lamp shall be out.
(g) Place switch "5" in the "down" position.
(h) Lamps "1" through "9" shall be out.

4.9.15 Mode control tests, involving the "M" flip-flop.

Step 1 (a) Push the "Power On" button if PH lamp is not lighted.
(b) Place the "Delay Start" switch in the "OFF" position.
(c) Set the "Operate" switch to the "RESET" position, then release to the "MAN" position.
(d) The operate lamp shall light.
(e) The PH lamp shall go out.
(f) Depress and release the "Power On" button.

Step 2 (a) Place the "Delay Start" switch in the "ON" position.
(b) Place the "OPERATE" switch in the "RESET" position, then release it to the "MAN" position.
(c) The "DS" lamp shall light.
(d) The "PH" lamp shall go out.

Step 3 (a) Place the "Delay Start" switch in the "OFF" position.
(b) Depress and release the "Power On" button.
(c) Set the "OPERATE" switch in the "RESET" position, and release it to the "MAN" position.
(d) The "OPERATE" lamp shall light.
(e) Measure the voltage at the ATR test jack, on the front panel. It shall be plus 6 ± 1 volt.
(f) Place the "OPERATE" switch in the "TAPE" position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 3 (Cont'd)

(g) The "OP" lamp shall stay lighted.
(h) The ATR test voltage shall be minus 6 ± 1 volt.
(i) Place the "OPERATE" switch in the "MAN" position.
(j) Place switch "5" "up"; it will lock.
(k) Place the "Hand Sprocket" switch in the "OPERATE" position. Release the "Hand Sprocket" switch.
(l) The ATR test voltage shall be plus 6 ± 1 volt.

4.9.16 Checkout of the "M" flip-flops for manual halt.

Step 1
(a) Place -12V ± IV on Pin J5-H. Connect the return to J5-V.
(b) Place the "OPERATE" switch in the "TAPE" position.
(c) The "OPERATE" lamps shall be lighted.
(d) Using the item 1 (VOM), measure the ATR voltage at Pin J5-W. As referred to power ground at Pin J8-K, the voltage read shall be -6 ± 1 volt.
(e) Using item 1 (VOM), measure the HTR voltage at Pin J5-X. The voltage read shall be plus 6 ± 1 volt.
(f) Place the "OPERATE" switch in the "MAN" position.
(g) The voltage at J5-W shall be minus 6 ± 1 volt.
(h) Depress and release switch "6".
(i) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(j) Measure the voltage at J5-W. It shall be +6 ± 1 volt.
(k) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(l) The voltage at J5-W shall be +6 ± 1 volt.
(m) The "OP" lamp shall be lighted.
(n) Repeat voltage measurement of 4.9.16e. The voltage shall be the same.

Step 2 Photoelectric tape reader halt at end of next marker.-

(a) Remove -12V from Pin J5-H. Place switch "5" "down".
(b) Depress and release switch "7". Lamp "7" shall light.
(c) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(d) The "Marker Selector" display shall read ##2.
(e) The OP lamp shall be out.
(f) The HS lamp shall be lighted.
(g) The "Command Indicator" ISF lamp shall be lighted.
(h) Depress and release switches "6" and "7".
(i) Lamps "6" and "7" shall light.
(j) Place the "Head Sprocket" switch in the "OPERATE" position and replace.
(k) The "Marker Selector" display shall read #23.
(i) Depress and release switch "8". Lamp "8" shall light.
(m) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(n) The "Marker Selector" display shall read 234.
(o) The HS and OP lamps shall not be lighted.
(p) The PH and "Command Indicator" ITA lamps shall be lighted.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the currentversion before use.

Step 2 (Cont'd)

(q) Using item (VOM), measure ATR voltage at Pin J5-W. The voltage read shall be plus 6 ± 1 volt.
(r) Using item (VOM), measure HTR voltage at Pin J5-X. The voltage read shall be -6 ± 1 volt.

4.9.17 Program skip test.

Step 1 (a) Place the "homing" switch in the "OFF" position.
(b) Depress the "Power On" button.
(c) Marker display shall red ###.
(d) Index display shall read 000.
(e) Place the "OPERATE" switch in the "RESET" position, and release.
(f) The OP lamp shall be lighted.

Step 2 (a) Depress and release switches "2", "4" and "6".
(b) Place switch "5" in "up" position; it shall lock.
(c) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(d) The ITF lamp shall light. The OP lamp shall remain lighted.

Step 3 (a) Depress and release switches "4" and "9".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(c) The BSC lamp shall light.

Step 4 (a) Depress and release switches "2", "3", "4" and "6".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(c) The ITF lamp shall light. The OP lamp shall remain lighted.

Step 5 (a) Depress and release switches "4" and "9".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(c) The BSC lamp shall light.

Step 6 (a) Depress and release switches "2" and "3" and "6".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(c) The ITF lamp shall light, and the OP lamp shall remain lighted.
(d) Depress and release switches "4" and "9".
(e) Place the "Hand Sprocket" switch in the "OPERATE" position, and release.
(f) The BSC lamp shall light.

Step 7 (a) Place the "Program Skip" switch "A" in the "ON" position.
(b) Depress and release switches "2" "4" and "6".
(c) Place the "Hand Sprocket" switch in the "OPERATE" position, and release.
(d) The "Program Skip" lamp "A" shall light.
(e) The "ISF" lamp shall light.
(f) Place the "Program Skip" switch "A" in the "OFF" position.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.

Step 7 (Cont'd)

(g) The ISF lamp shall light. The "Program Skip" lamp "A" shall go out.
(h) Place the "Program Skip" switch "A" in the "ON" position.
(i) Depress and release switches "2", "4" and "6".
(j) Place the "Hand Sprocket" switch in the "OPERATE" position, and release.
(k) The ISF lamp shall light.
(l) "Program Skip" lamp "A" shall light.

Step 8 (a) Place the "Program Skip" switch "B" in the "ON" position.
(b) Depress and release switches "2", "3", "4", and "6".
(c) Place the "Hand Sprocket" switch in the "OPERATE" position, and release.
(d) No command light shall light except 18 VFT.
(e) The "Program Skip" lamp "B" shall light.
(f) Place the "Program Skip" switch "B" in the "OFF" position.
(g) The ISF lamp shall light.
(h) The "Program Skip" lamp "B" shall go out.
(i) Place the "Program Skip" switch "B" in the "ON" position.
(j) Depress and release switches "2", "3", "4", and "6".
(k) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(l) No command light except 18 VFT shall be on.
(m) The "Program Skip" lamp "B" shall light. "A" shall remain lighted.

Step 9 (a) Place the "Program Skip" switch "C" in the "ON" position.
(b) Depress and release switches "2", "3", and "6".
(c) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(d) The BSC lamp shall light.
(e) The "Program Skip" lamp "C" shall light.
(f) Place the "Program Skip" switch "C" in the "OFF" position.
(g) No command light except 18VFT shall be on.
(h) Place the "Program Skip" switch "C" in the "ON" position.
(i) Depress and release switches "2", "3" and "6".
(j) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(k) The BSC lamp shall light. The "Program Skip" "C" shall light.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 10
(a) Depress and release switches "1", "4" and "6".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(c) The OTA lamps shall light.
(d) The "Program Skip" lamp "A" shall go out.
(e) Depress and release switches "1", "3", "4" and "6".
(f) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(g) The OTF lamp shall light.
(h) The "Program Skip" lamp "B" shall go out.
(i) Depress and release switches "1", "3" and "6".
(j) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(k) The ITF lamp shall light. The DS lamp shall light.
(l) The "Program Skip" lamp "C" shall go out.
(m) Depress and release switches "1" and "9".
(n) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(o) The OP lamp shall light. The OSF lamp shall light.
(p) The DS lamp shall go out.
(q) Depress and release switches "2", "3", "4" and "6".
(r) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(s) The ITA lamp shall light. The "Program Skip" lamp "B" shall light.
(t) The OP lamp shall go out.

Step 11
(a) Place the "Program Skip" switch "B" in the "OFF" position.
(b) Place the "OPERATE" switch in the "RESET" position and then in the "MAN" position.
(c) The OP lamp shall light.
(d) Depress and release switches "2", "3" and "6".
(e) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(f) The "Program Skip" lamp "C" shall light. The OTF lamp shall light.
(g) The OP lamp shall go out.
(h) Place the "Program Skip" switches "A" and "C" in the "OFF" position.
(i) Place the "OPERATE" switch in the "RESET" position, and then in the "MAN" position.

4.9.18 Jump command test.

Step 1
(a) Depress and release switches "1", "2", "3" and "6". Switch "5" shall be in the "up" position.
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(c) The HS lamp shall light. The OP lamp shall go out.

Step 2
(a) Place switch "5" down.
(b) Place the "Test Mode" switch in the "STATIC" position.
(c) The "Test Normal" lamp shall go out.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 2 (Cont'd)
(d) Place -12V on Pin J5-H.
(e) Depress and hold switch "1".
(f) Operate the "Clock" key once.
(g) Release switch "1".
(h) Lamp "1" shall light.
(i) Depress and hold switch "2".
(j) Operate the "Clock" key once.
(k) Release switch "2".
(l) Depress and hold switch "6".
(m) Operate the "Clock" key once.
(n) Release switch "6".
(o) Place the "Operate" switch in the "TAPE" position.
(p) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(q) Operate the "Clock" key twice.
(r) Release the "Hand Sprocket" switch.
(s) Lamps "1", "2", P1, and P2 shall be lighted.
(t) Operate the "Clock" key six (6) times.
(u) The "HS" lamp shall go out. The "OP" lamp shall light.
(v) Operate the "Clock" key twice.
(w) Lamps "1" and "2" shall go out.
(x) Remove -12V from Pin J5-H.
(y) Place the "Test Mode" switch in the "NOR" position.
(z) The "Test Normal" lamp shall light.
(a) Place switch "5" in the "up" position.
(b) Place the "Operate" switch in the "RESET" position, and release to the "MAN" position.

4.9.19 Command halt test.

Step 1
(a) The "Preset Stop" switch shall remain in the "NORM" position.
(b) Depress and release switches "4", "2", "1" and "6".
(c) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(d) The PH lamp shall be lighted.
(e) The "CH" lamp shall be lighted.
(f) Lamps "1", "2", "4" and "5" shall be lighted.

Step 2
(a) Set the "Index Selector" switches to 199.
(b) Place the "Preset Stop" switch in the "INDEX" position.
(c) Place the "OPERATE" switch in the "RESET" position, and then in the "MAN" position.
(d) Depress and release switches "1", "2", "4" and "6".
(e) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(f) The HS lamp shall be lighted.
(g) The CH, PH and OP lamps shall not be lighted.

Step 3
(a) Place the "OPERATE" switch in the "RESET" position.
(b) Place the "Preset Stop" switch in the "Clock" position.
(c) Depress and release switches "1", "2", "4" and "6".
(d) Place the "Hand Sprocket" switch in the "OPERATE" position and release.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.

Step 3 (Cont'd)
(e) The "OP" lamp shall remain lighted.
(f) The "CH" and "PH" lamps shall not be lighted.

4.9.20 Preset stop at clock test.

Step 1
(a) Depress and release switches "3", "4" and "9".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release. The "18VFT" lamp shall not light.
(c) Depress and release switch "9" and "1".
(d) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(e) The "PH" lamp shall light.
(f) Place the "OPERATE" switch in the "RESET" position, and then in the "MAN" position.
(g) The "OP" lamp shall light.
(h) Depress and release switches "1", "3", "4" and "9".
(i) Place the "Hand Sprocket" switch in the "OPERATE" position and release. The "18VFT" lamp shall light.

4.9.21 Preset stop at index test.

Step 1
(a) Depress and release the "Power On" button.
(b) Place the "OPERATE" switch in the "RESET" position, then in the "MAN" position.
(c) Place the "Preset Stop" switch in the "INDEX" position. Place the Homing Selector in Marker/Index.
(d) Set the "Index Selector" rotary switches to 012.
(e) Switch "5" shall remain in the "up" position.
(f) Place "Hand Sprocket" switch in the "OPERATE" position and release.
(g) The "Index Selector" display shall read 001.
(h) Repeat Step 1f eleven times.
(i) The "Index Selector" display shall read 012.
(j) The PH lamp shall not be lighted.
(k) Repeat Step 1f once.

Step 2
(a) The PH lamp shall light.
(b) The "Index Selector" display shall read 013.
(c) Repeat Step 1f once.
(d) The "Index Selector" display shall continue to read 013.
(e) Place the "OPERATE" switch in the "RESET" position, and then in the "MAN" position.
(f) Place the "Preset Stop" switch in the "NORM" position.

4.9.22 Command control switch test.

Refer to Table XX for each line:

(a) Place the "Hand Sprocket" switch in the "CLEAR" position, and release, as indicated in column 1.
(b) Place the appropriate "Command Control" switch in the position listed in column 2 and release.
(c) Lamps as listed in column 3 shall be lighted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

TABLE XX

1	2	3
Place "Hand Sprocket" in position shown below. Then release "Hand Sprocket" switch.	Place the appropriate "Command Control" switch in the position listed below, and then release switch.	The following lamps shall be lighted:
CLEAR	ISA	9,5,1
CLEAR	OSA	9,5
CLEAR	ISF	9,5,2,1
CLEAR	OSF	9,5,2
CLEAR	OTF	9,5,3,2
CLEAR	ITF	9,5,3,2,1
CLEAR	OTA	9,5,4,3,2
CLEAR	ITA	9,5,4,3,2,1
CLEAR	BSC	9,5,4,2
CLEAR	-18VFT ON	9,5,4,3,1
CLEAR	-18VFT OFF	9,5,4,3
CLEAR	NONE	5

4.9.23 O Flip-Flop command loading test.

Step 1
(a) Refer to Table XXI. Place the appropriate "Command Control" switch in the position listed in column 1, and release.
(b) Place the "Hand Sprocket" switch in the "OPERATE" position, and release as indicated in column 2.
(c) The lamp listed in column 3 shall light.
(d) After completion of Table XXI operation, depress the -18VFT OFF switch. Place the "Hand Sprocket" switch in operate and release.

TABLE XXI

1	2	3
Place the appropriate "Command Control" switch in the position listed below and then release switch	Place the "Hand Sprocket" switch in the position listed below, and then release switch.	The following lamps shall be lighted.
ITF	OPERATE	ITF
OTF	OPERATE	OTF
ISF	OPERATE	ISF
OSF	OPERATE	OSF
ITA	OPERATE	ITA
OTA	OPERATE	OTA
OSA	OPERATE	ISF
ISA	OPERATE	OSF
BSC	OPERATE	BSC
-18VFT ON	OPERATE	-18VFT

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 2
(a) Depress and release switches "4" and "9".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(c) The BSC lamp shall light.
(d) The -18V FT lamp shall not light.

Step 3
(a) Depress and release switches "1", "2" and "9".
(b) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(c) No "Command Indicator" lamp shall light, except -18VFT and OTF.

TABLE XXII

D.C. Supply	ØA	ØB	ØC	VAB,BC,CA(RMS Volt) Max	Min
-6	319A - 52	319A - 46	319A - 41	14.0	13.4
-18	321A - 52	321A - 46	321A - 41	32.2	31.0
-10	321A - 11	321B - 11	321A - 1	17.0	16.4
+6	321B - 62	321B - 33	321A - 33	16.5	15.9
+35	319B - 62	319B - 33	319A - 33	46.0	44.0
-14	315B - 52	315B - 60	315B - 63	11.6	11.2
+15	321A - 24	321B - 20	321A - 17	22.6	21.8
+150	317A - 36	317A - 38	317A - 40	184	180
+150-1	315A - 36	315A - 38	315A - 40	184	180
-32	319A - 24	319B - 20	319A - 17	25.0	24.0
-27	319A - 11	319B - 11	319A - 1	21.8	20.8
+20	223A - 22	223B - 17	223A - 19	16.4	15.6

4.9.24 Error detector tests.

Step 1
(a) Place the "Operate" switch in the "RESET" position and then in the "MAN position.
(b) The SFC switch shall be in the "NORMAL" position.
(c) Switch "5" shall be in the "up" position.
(d) Depress the OSA switch and release. Operate the "Hand Sprocket" switch.
(e) Place the appropriate "Command Control" switch in the OTF position and release.
(f) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(g) Depress and release switches "1", "3" and "6".
(h) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(i) The ERROR lamp on the front of the FALT shall not "flash".
(j) The OP lamp shall be lighted.

Step 2
(a) Place the "Operate" switch in the "RESET" position and release.
(b) Place the "SFC" switch located on front panel of FALT to "TEST" position.
(c) The Test Normal lamp shall go out.
(d) Place the HF-OTF "Command Control" switch in the OTF position and release.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.

Step 2 (Cont'd)
(e) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(f) Depress and release switches "1", "3", "4", "7" and "9".
(g) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(h) The ERROR lamp shall "flash".
(i) The OP lamp shall go out.
(j) The F1 lamp shall be lighted.
(k) The FO lamp shall not be lighted.

Step 3
(a) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(b) Place the "SFC" switch to "NOR" position.
(c) The Test Normal lamp shall be lighted.
(d) Place the ITF-OTF "Command Control" switch in the ITF position and release.
(e) Place the "Hand Sprocket" switch in the "OPERATE" position and then release.
(f) Depress and release switches 1, 2, 6 and 9.
(g) Place the "Hand Sprocket" switch in the "OPERATE" position and release.
(h) The ERROR lamp shall "flash".
(i) The OP lamp shall go out.
(j) The F1 lamp shall not be lighted.
(k) The FO lamp shall not be lighted.

NOTE: Hereinafter, "Operate the "Hand Sprocket" switch" shall be interpreted to mean: Place the "Hand Sprocket" switch in the "OPERATE" position, and then release the "Hand Sprocket" switch.

Step 4
(a) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(b) Place the ITA-OTA "Command Control" switch in the "OTA" position and release.
(c) Operate the "Hand Sprocket" switch.
(d) Depress and release switch "6", "7", "1" and "3".
(e) Operate the "Hand Sprocket" switch.
(f) The ERROR lamp shall "flash".

Step 5
(a) Place the "Operate" switch in the "RESET" position and release.
(b) Place the ITA-OTA "Command Control" switch in the "ITA" position.
(c) Operate the "Hand Sprocket" switch.
(d) Depress and release switch "6", "7" and "1".
(e) Operate the "Hand Sprocket" switch.
(f) The ERROR lamp shall "flash".
(g) Place the "Operate" switch in the "RESET" position and release.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.25 Matrix driver output test.*- Apply the load of Figure 3 to each of the pins of Tables XXIII and XXIV simultaneously.

Step 1
(a) Place the "Operate" switch in the "RESET" position and then in the "MAN" position.
(b) The OP lamp shall be lighted.
(c) Using item 1 (VOM) measure the voltage at Pins J4-A, B, C, D, E, F, G, H,J,K,L,M,P,R,S,T,U,V,W,X,Y and Z with respect to power ground. The voltage shall be -.15 ± .15.
(d) Measure the voltage at Pins J4-a, b, c, d, e, f, g and h. The voltage shall be +2.85 ± .15 volts. Measure the voltage at J4-N. It shall be minus 6 ± 1 volt.

Step 2
(a) Place the "Hand Sprocket" switch in the "CLEAR" position and release.
(b) Depress and release switch "9". Lamp "9" shall light.
(c) Place the "Test Mode" switch in the "STATIC" position.
(d) The Test Normal lamp shall go out.
(e) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.

*NOTE: The following procedure may be used in place of the voltage measurements described in this section:
Connect indicator lights between each of the driver outputs of Step 1 - c, d and e, and ground. The lights shall work at the voltage levels specified, turning on or off with a change in state of the driver being monitored. Make an initial measurement of the voltage extremes encountered at the pins of 1c, 1d, and 1e during circuit operation. After correct light operation is assured, the indicator lights may be used to denote the state of the driver.

(f) Operate the "Clock" key twice.
(g) Release the "Hand Sprocket" switch.
(h) Operate the "Clock" key three (3) times.
(i) The "OP" lamp shall not be lighted. Lamps P2 and P4 shall be lighted.
(j) The DC voltage at each of the Pins listed in Step 1c, and Step 1d, shall be $-6^{+1}_{-1.8}$ volts.

Step 3
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Place the "Hand Sprocket" switch to the "CLEAR" position, and release.
(c) Depress and release the "Power On" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) The "OP" lamp shall be lighted.
(f) Depress and release switches "4" and "8".

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

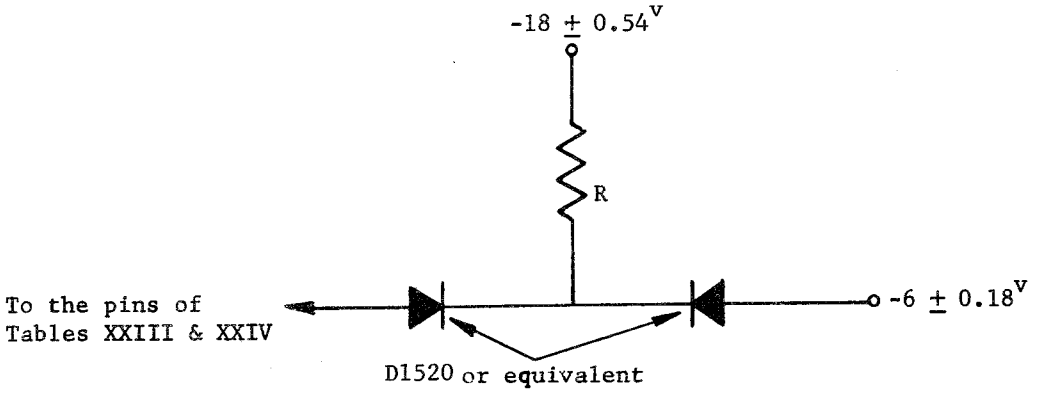
TABLE XXIII

Logic Driver Pins J4-	R Loads to Each Pin	DU	DL	Control Box Switches R-
A	120 Ohms, 5%, 10W	02	--	7
B	"	03	--	67
C	"	04	--	8
D	"	05	--	68
E	"	06	--	78
F	"	07	--	678
G	"	11	--	69
H	"	12	--	79
J	"	13	--	679
K	"	14	--	89
L	"	15	--	689
M	"	16	--	789
N	"	--	20	5
P	"	--	21	15
R	"	--	22	25
S	"	--	23	125
T	"	--	24	35
U	"	--	25	135
V	"	--	26	235
W	"	--	27	1235
X	"	--	30	45
Y	"	--	31	145
Z	"	--	32	245

TABLE XXIV

Logic Driver Pins J4-	R Loads to Each Pin	DUR	DLR	Control Box Switches R-
a	1.2K Ohms, 5%, 1W	02	--	7
b	"	03	--	67
c	"	04	--	8
d	"	05	--	68
e	"	--	33	1245
f	"	--	34	345
g	"	--	35	1345
h	"	--	36	2345

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.



LOAD FOR LOGIC DRIVERS

FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.25 Step 3 Cont'd.

(g) Place the "Test Mode" switch in the "STATIC" position.
(h) Place and hold the "Hand Sprocket" switch in "OPERATE" position.
(i) Operate the "Clock" key twice.
(j) Release the "Hand Sprocket" switch. Lamps P1, P2, and OP, shall be lighted.
(k) Operate the "Clock" key four (4) times.
(l) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(m) The voltage measured at Pins J4-d, J -C, and J -X shall be -6 ± 1 volts.
(n) The voltage at all other pins listed in Step 1c. shall be -.15 ± .15 volts. The voltage at all other pins listed in Step 1d. except c shall be 2.85 ± .15 volts.

Step 4
(a) Repeat Steps 3.a. through 3.e.
(b) Depress and release switches "7" and "3".
(c) Place the "Test Mode" switch in the "STATIC" position.
(d) Place and hold the "Hand Sprocket" switch in "OPERATE" position.
(e) Operate the "Clock" key twice.
(f) Release the "Hand Sprocket" switch.
(g) Operate the "Clock" key four (4) times.
(h) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(i) The voltage measured at Pins J4-A, J4-a, and J4-T shall be -6 ± 1 volts.
(j) Repeat Step 3n.

Step 5
(a) Repeat Steps 3a, through 3e.
(b) Depress and release switches "2", "6", and "7".
(c) Place the "Test Mode" switch in the "STATIC" position.
(d) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(e) Operate the "Clock" key twice.
(f) Release the "Hand Sprocket" switch.
(g) Operate the "Clock" key four (4) times.
(h) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(i) The voltage measured at Pins J4-B, J4-b, and J4-R shall be -6 ± 1 volts.
(j) Repeat Step 3n.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 6
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power On" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "6" and "8".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch.
(j) Operate the "Clock" key four (4) times.
(k) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(l) The voltage measured at Pins J4-D, J4-d, and J4-N shall be -6 ± 1 volts.
(m) Repeat Step 3n.

Step 7
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power On" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "1", "8", and "7".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch.
(j) Operate the "Clock" key four (4) times.
(k) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(l) The voltage measured at Pins J4-E and J4-P shall be -6 ± 1 volts.
(m) Repeat Step 3n.

Step 8
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power On" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "1", "2", "6", "7" and "8".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch and operate the "Clock" key four (4) times.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

(j) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(k) The voltage measured at Pins J4-E and J4-S shall be -6 ± 1 volts.
(l) Repeat Step 3n.

Step 9
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power On" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "1", "3", "6", and "9".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch and operate the "Clock" key four (4) times.
(j) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(k) The voltage measured at Pins J4-G and J4-U shall be -6 ± 1 volts.
(l) Repeat Step 3n.

Step 10
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power On" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "2", "3", "7" and "9".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch and operate the "Clock" key four (4) times.
(j) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(k) The voltage measured at Pins J4-H and J4-V shall be -6 ± 1 volts.
(l) Repeat Step 3n.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 11
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power on" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "1", "2", "3", "6", "7", and "9".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch and operate the "Clock" key four (4) times.
(j) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(k) The voltage measured at Pins J4-J and J4-W shall be -6 $\pm$ 1 volts.
(l) Repeat Step 3n.

Step 12
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power ON" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "1", "4", "8", and "9".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch and operate the "Clock" key four (4) times.
(j) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(k) The voltage measured at Pins J4-K and J4-Y shall be -6 $\pm$ 1 volts.
(l) Repeat Step 3n.

Step 13
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power ON" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "2", "4", "6", "8", and "9".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch and operate the "Clock" key four (4) times.
(j) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(k) The voltage measured at Pins J4-L and J4-Z shall be -6 $\pm$ 1 volts.
(l) Repeat Step 3n.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 14
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power ON" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "1", "2", "4", "7", "8", and "9".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch and operate the "Clock" key four (4) times.
(j) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(k) The voltage measured at Pins J4-M and J4-e shall be -6 $\pm$ 1 volts.
(l) Repeat Step 3n.

Step 15
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power ON" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "3", "4", "6", "8", and "9".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch and operate the "Clock" key four (4) times.
(j) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(k) The voltage measured at Pins J4-f and J4-L shall be -6 $\pm$ 1 volts.
(l) Repeat Step 3n.

Step 16
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power ON" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "1", "3", "4", and "6", "7", "8" and "9".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch and operate the "Clock" key four (4) times.
(j) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(k) The voltage measured on Pin J4-g shall be -6 $\pm$ 1 volts.
(l) Repeat Step 3n.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 17
(a) Place the "Test Mode" switch in the "NOR" position.
(b) Momentarily place the "Hand Sprocket" switch in the "CLEAR" position.
(c) Depress and release the "Power ON" button.
(d) Place the "Operate" switch in the "RESET" position, and then in the "MAN" position.
(e) Depress and release switches "2", "3", "4", and "6", "7", "8", and "9".
(f) Place the "Test Mode" switch in the "STATIC" position.
(g) Place and hold the "Hand Sprocket" switch in the "OPERATE" position.
(h) Operate the "Clock" key twice.
(i) Release the "Hand Sprocket" switch and operate the "Clock" key four (4) times.
(j) Lamps OP, P1, P3, and P4 shall not be lighted. Lamp P2 shall be lighted.
(k) The voltage measured on pin J4-h shall be -6±1 volts.
(l) Repeat step 3n.

4.9.26 FIT and FOT sense lines.- This test shall be conducted using OTF and ITF commands.

Step 1 Connect the FD839 output lead of a crosspoint circuit as shown in Figure 4 to FIT at pin J13-i. Connect a similar circuit to FOT at pin J13-j.

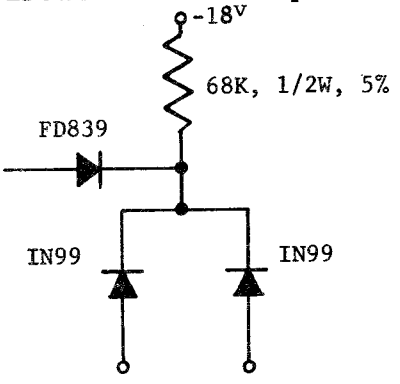


FIGURE 4

Step 2 To the IN99 inputs of the FIT crosspoint, connect DU02, pin J4-A and DL20, pin J4-N. To the IN99 input of the FOT crosspoint, connect DU02, pin J4-A, and DL21, pin J4-P. Using the following procedure, test the FIT and FOT lines:

(a) Place the "Test Mode" switch in "NORMAL" position.
(b) Place the "Hand Sprocket" switch in the "CLEAR" position, and release.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Step 2 Cont'd

(c) Place the "Operate" switch in the "RESET" position, and release.
(d) Place switch "5" in the "up" position.
(e) The "Operate" lamp and lamp "5" shall be "on". Lamps "1" through "4", and "6" through "9" shall be "off".
(f) Place the appropriate command switch to the OTF position, and then release. Place the "Hand Sprocket" switch in the "OPERATE" position. The OTF lamp shall light.
(g) Depress switch "7", and release. Lamp "7" shall light.
(h) Depress switch "1", and release. Lamp "1" shall light.
(i) Depress the "Hand Sprocket" switch to the "OPERATE" position, and then release. The ERROR lamp shall not flash and the "Operate" lamp shall remain on.
(j) Depress switch "7", and release.
(k) Depress the "Hand Sprocket" switch to the "OPERATE" position, and then release. The ERROR lamp shall "flash" and the operate lamp shall go out.
(l) Place the "Operate" switch in the "RESET" position, and release.
(m) Place the appropriate command switch to the ITF position, and then release. Place the "Hand Sprocket" switch in the "OPERATE" position. The ITF lamp shall light.
(n) Depress switch "7", and release.
(o) Depress the "Hand Sprocket" switch to the "OPERATE" position, and then release. The ERROR lamp shall not "flash" and the operate lamp shall remain "on".
(p) Depress switch "7", and release. Depress switch "1", and release.
(q) Depress the "Hand Sprocket" switch to the "OPERATE" position, and then release. The ERROR lamp shall flash and the operate lamp shall go out.
(r) Place the "OPERATE" switch in the "RESET" position, and release. The ERROR lamp shall stop flashing and the "Operate" lamp shall light.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.9.27 One-set and zero-set test of E18.

Step 1 (a) Place "Operate" switch in "RESET" position, and then release.
(b) Place switch "5" in "up" position.
(c) Depress switches "3", "4", and "9".
(d) Operate the "Hand Sprocket" switch.
(e) The "18 VFT" Lamp shall not light.
(f) With Item 1 (VOM) measure the voltage from pin J4-PP to ground. This voltage shall read 0 +IV volt. -0V
(g) Place the "Hand Sprocket" swtich in the "CLEAR" position, and release.
(h) Depress switches "1", "3", "4", and "9".
(i) Operate the "Hand Sprocket" switch.
(j) The "18 VFT" Lamp shall light.
(k) Measure the voltage from pin J4-PP to ground. This voltage shall read between -17.46 and -18.54.

4.9.28 Self-test.- This test shall be conducted using self-test tape with instructions on Drawing B8623275 and Signal Data Reproducer F10531830. This tape shall be run with the voltage marginal switch in the OFF position first, then in each of the other five positions.

4.10 FALT run-in.- Subsequent to the successful performance of 4.9 of this specification, the FALT shall be subjected to a 6-1/2 hour run-in time. Run-in shall exclude any down-time necessary for repairs. It shall include the 1/2 hour vibration of 4.10.1, and following vibration at least 2 hours operation at -22°F, at least 2 hours operation at +125°F, and at least 2 hours operation at nominal room temperature. (Refer to 4.10.2).

4.10.1 Vibration tests.- The FALT, in its transit case, shall be vibrated at 271 ± 2 cycles per minute with a circular motion of 1.0 ± .1 inch diameter. The horizontal motion shall be constrained to a minimum of 3.25 inches total excursion. The unit shall be vibrated for 15 minutes in its normal operating position, and then vibrated for 15 minutes in an up-side-down position. The FALT shall then meet the tests of 4.10.2.

4.10.2 Temperature run-in.

(a) After completing the vibration of 4.10.1, the FALT shall be placed in a chamber and the temperature raised at a rate not to exceed 2 degrees per minute to +125°F. After temperature stabilization is reached, the FALT self-test tape shall be run continuously for at least 2 hours.

(b) Repeat (a) except the temperature shall be -22°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

(c) Following the successful completion of (a) and (b), perform the tests of 4.9 at a temperature between +55°F and +95°F for at least 2 hours, or the time required to complete 4.9, whichever is greater.

NOTE: Run-in failures may be corrected and the tests continued without redoing that portion of the tests already acceptable with the following provisions:

(a) Any part, module, or subassembly that is used as a replacement shall have previously passed those environmental tests which the FALT has already passed.

(b) Any replacement part shall have been submitted, for 1 hour, to the temperature at which FALT has been accepted.

(c) After correction, at least one complete test sequence is accomplished at the temperature at which failure occurred.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be as determined by the Government.

6. NOTES

6.1 Intended use.- The Logic Unit Test Set Computer: 8623272 is commonly referred to as FALT. FALT is a rugged piece of portable test equipment used to maintain Gun Direction Computer M18 and associated equipment in the field and depot. FALT is essentially a control center and a multiple electronic switch with the ability to single out and select any flip-flop in the computer through an addressing code punched paper tape. Preprogrammed paper tapes are prepared which instruct FALT, through a signal data reproducer, to direct the state of the computer flip-flops, apply single clock pulses, and sample the results. When an error is detected, FALT stops the operations, flashes a trouble light, and displays the number of the test being performed. The operator can isolate the board(s) that should be replaced. FALT maintenance of computer direction equipment has been reduced essentially to the circuit board(s) replacement level.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Applicable stock number

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1220-A205Y

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.